Translation
(Protein Synthesis)

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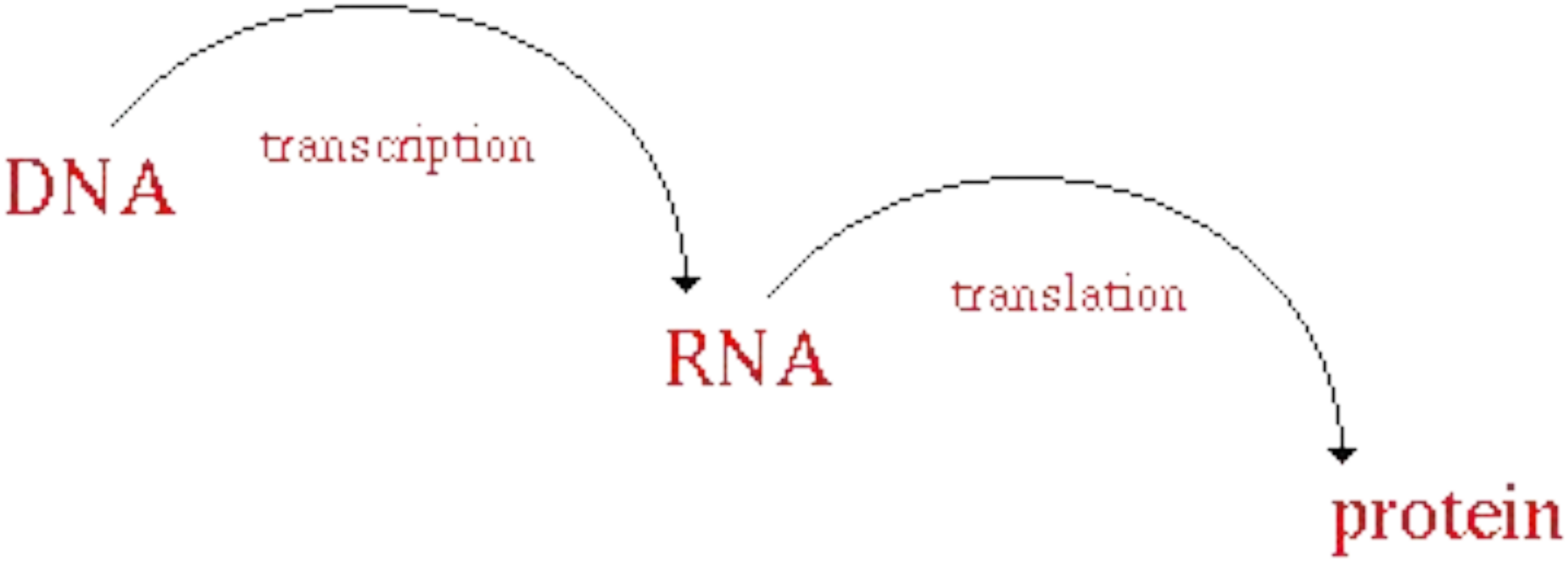
Objectives:

At the end of this chapter, it is expected;

- To describe the characteristics of genetic code, codon, and anticodon
- To describe the features of translation.
- To identify the essential components involve in translation.
- To explain the mechanism of protein synthesis that occurs on RNA-protein complexes termed ribosomes.
- To appreciate that protein synthesis is precisely controlled through the action of multiple accessory factors that are responsive to multiple extra- and intracellular regulatory signaling inputs.
- To explain the mechanism of posttranslational modification.
- To appreciate the process of protein targeting and degradation.
- To describe the differences between translation in prokaryotes and eukaryotes.
- To identify the inhibitors of translation.

Protein Synthesis takes place in 2 stages



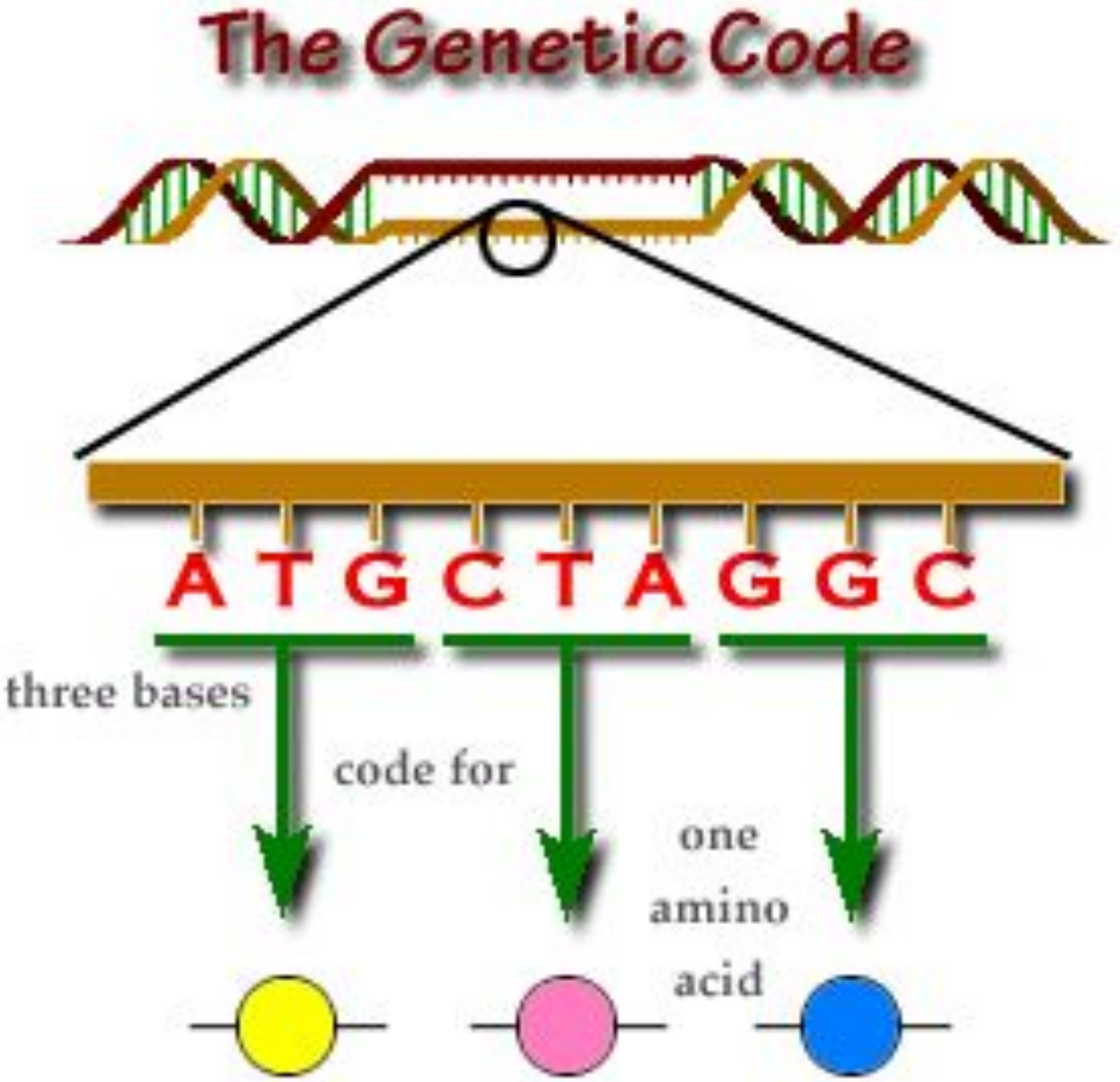
- DNA carries the code for every protein that can be made by a cell.
- A gene is a length of DNA which codes for a particular protein
- Transcription is the transfer of information from DNA to mRNA.
- Translation is the synthesis of protein based on a sequence specified by mRNA.

GENETIC CODE

- The genetic code - the set of rules – encoded in genetic material (DNA or RNA sequences) translated into proteins (amino acid sequences)
- Composed of four nucleotide bases i.e., A, G, C & U
- Tri-nucleotide sequences in mRNA called ‘Codons’ - determine the sequence of amino acids in the proteins
- Out of 64 codons, 61 codons code for 20 amino acids found in protein
- Written from 5’ to 3’ direction in mRNA.

Discovery:

- Philip Leder and Marshall Nirenberg were able to determine the sequences of 54 out of 64 codons
- Har Gobind Khorana identified the rest of the genetic code



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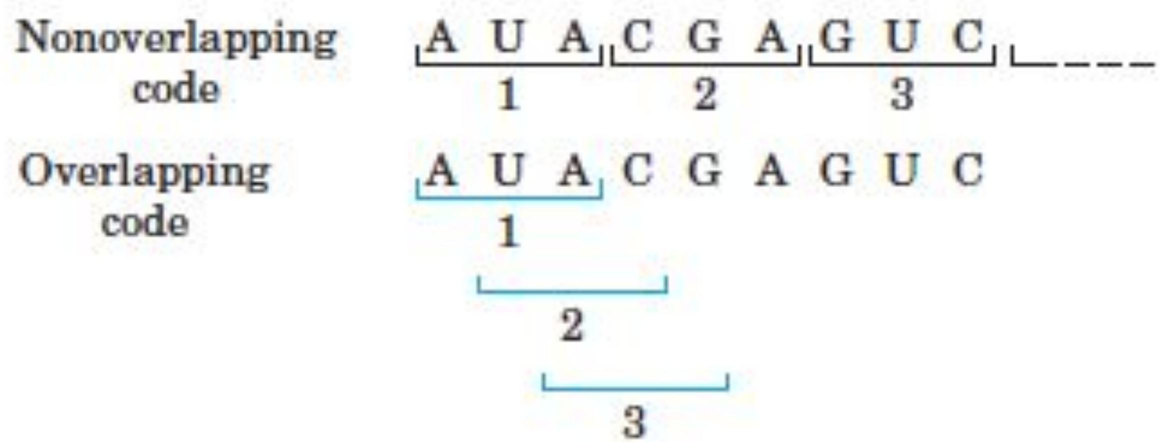
Characteristics of genetic code:

1. **Universal –** same codon used to code for same amino acids in all the living organisms (some exception, AUA – code for met in mitochondria, but isoleusine in cytoplasm)

2. **Specific –** A particular codon always codes for same aminoacids – highly specific (Unambiguous), e.g. UGG – code for tryp.

3. **Non-overlapping –** read from a fixed point, non-overlapping, commaless, and without punctuations. E.g. UUUCUUAGAGGG is read as UUU/CUU/AGA/GGG. Addition or deletion will radically change the message sequence.



Characteristics of genetic code:

4. **Degenerate** – most of the amino acids have >1 codons b/c 61 codons available to code for only 20 amino acids. E.g. glycine has four codons.

Wobble hypothesis explains codon degeneracy.

Degeneracy of the Genetic Code

Amino acid	Number of codons	Amino acid	Number of codons
Met	1	Tyr	2
Trp	1	Ile	3
Asn	2	Ala	4
Asp	2	Gly	4
Cys	2	Pro	4
Gln	2	Thr	4
Glu	2	Val	4
His	2	Arg	6
Lys	2	Leu	6
Phe	2	Ser	6

5. **Not punctuated** – there is no punctuation between codons, and the message is read in a continuing sequence of nucleotide triplets until a translation stop codon is reached.

Start/stop codons:

- Start/stop codons:
  - Three codons UAA, UAG & UGA – do not code for aminoacids, act as stop signal in protein synthesis – known as termination Or stop codons or nonsense codons.

    UAG – amber codon

    UAA – Ochre codon

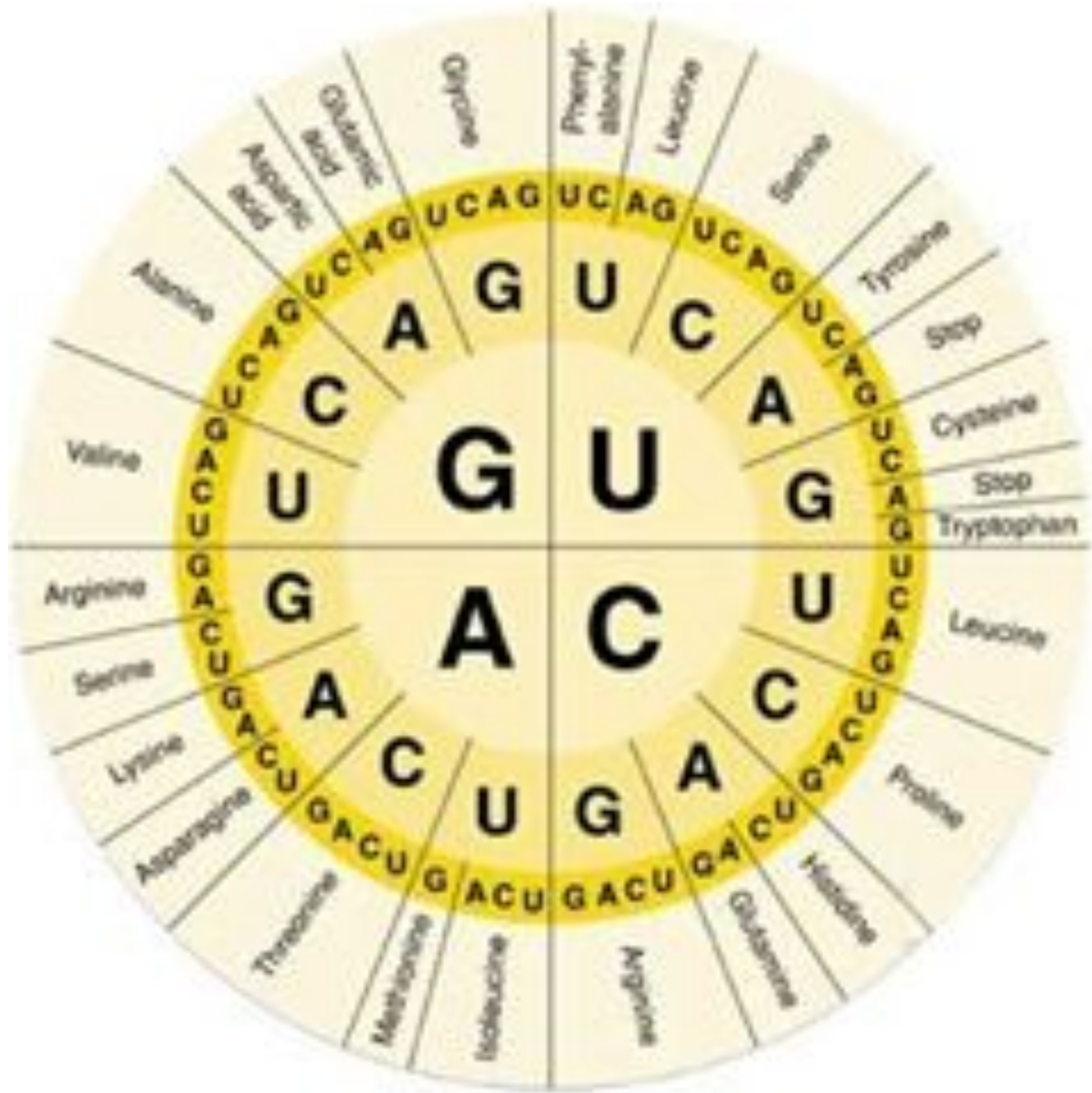
    UGA – opal codon
  - The Codon AUG and, sometimes GUG – are the chain initiation codons.

mRNA codons for Amino Acids

Second base

First base	U	C	A	G	Third base
U	UUU, UUC – Phenyl-alanine F UUA, UUG – Leucine L	UCU, UCC, UCA, UCG – Serine S	UAU, UAC – Tyrosine Y UAA Stop codon UAG Stop codon	UGU, UGC – Cysteine C UGA Stop codon UGG Tryptophan W	U C A G
C	CUU, CUC, CUA, CUG – Leucine L	CCU, CCC, CCA, CCG – Proline P	CAU, CAC – Histidine H CAA, CAG – Glutamine Q	CGU, CGC, CGA, CGG – Arginine R	U C A G
A	AUU, AUC, AUA – Isoleucine I AUG M Methionine start codon	ACU, ACC, ACA, ACG – Threonine T	AAU, AAC – Asparagine N AAA, AAG – Lysine K	AGU, AGC – Serine S AGA, AGG – Arginine R	U C A G
G	GUU, GUC, GUA, GUG – Valine V	GCU, GCC, GCA, GCG – Alanine A	GAU, GAC – Aspartic acid D GAA, GAG – Glutamic acid E	GGU, GGC, GGA, GGG – Glycine G	U C A G

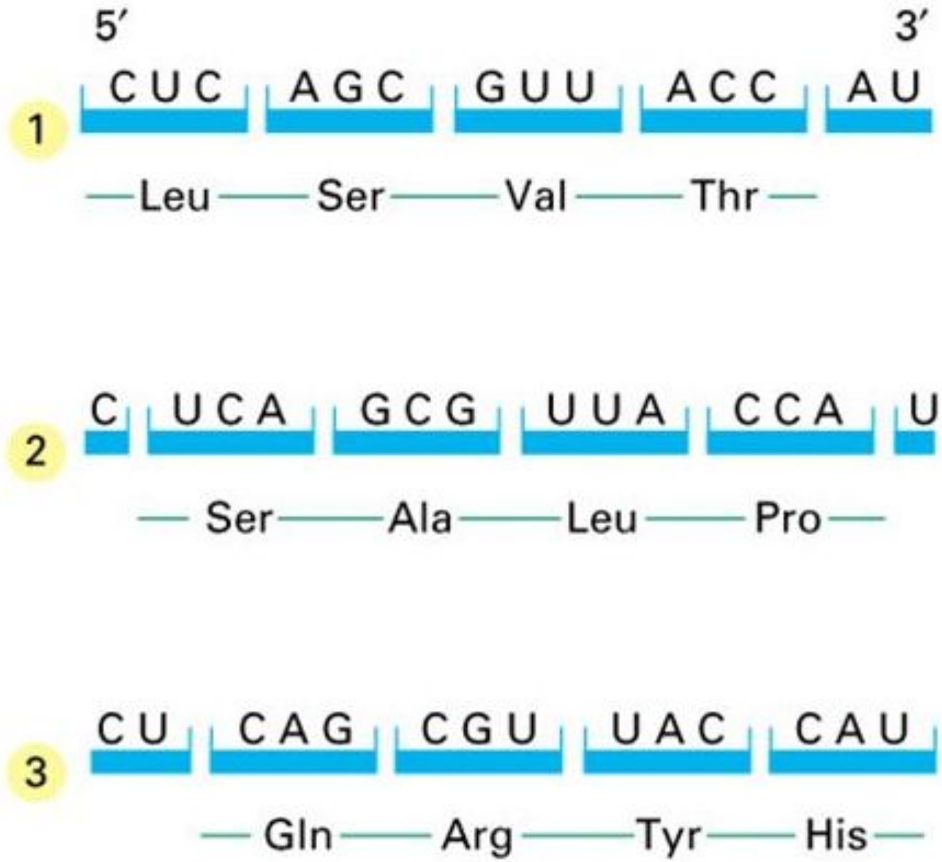
mRNA codons for Amino Acids



Reading frame:

- A codon is defined by the initial nucleotide from which translation starts
- Sets the frame for a run of uninterrupted triplets, which is known as an "open reading frame" (ORF).
- For example, the string “CUCAGCGUUACCAU..”
  - if read from the first position, the codons CUC, AGC, GUU, ACC, AU..
  - if read from the second position, the codons UCA, GCG, UUA, CCA, U..
  - if read starting from the third position, CAG, CGU, UAC, CAU..
- Every sequence can, thus, be read in its 5' → 3' direction
- With double-stranded DNA, there are six possible reading frames, three in the forward orientation on one strand and three reverse on the opposite strand

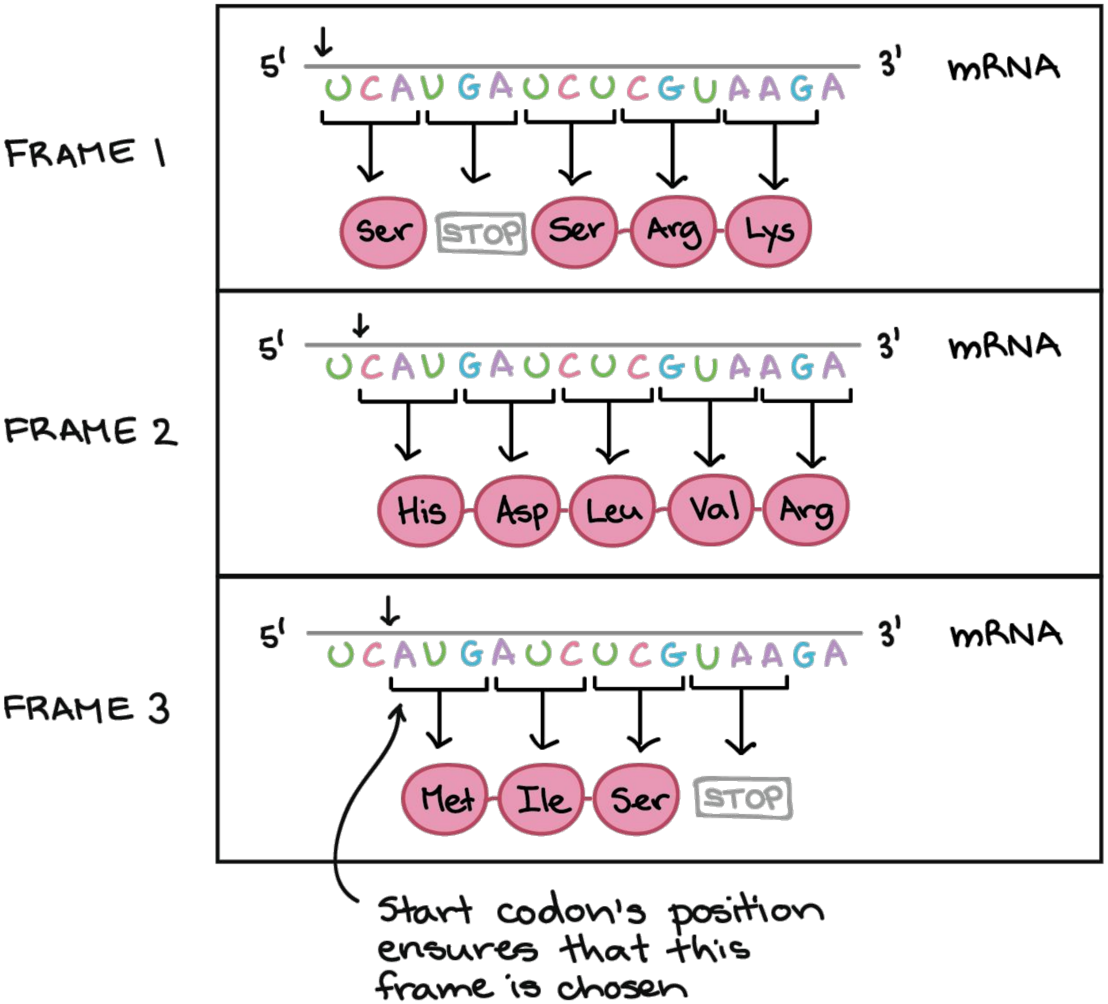
Reading frame:



- One of three possible reading frames contains protein message

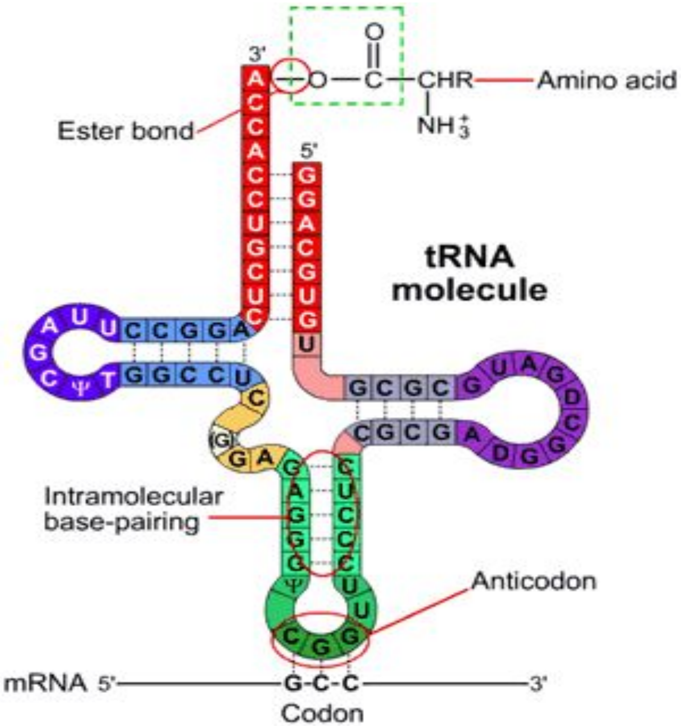
Reading frame:

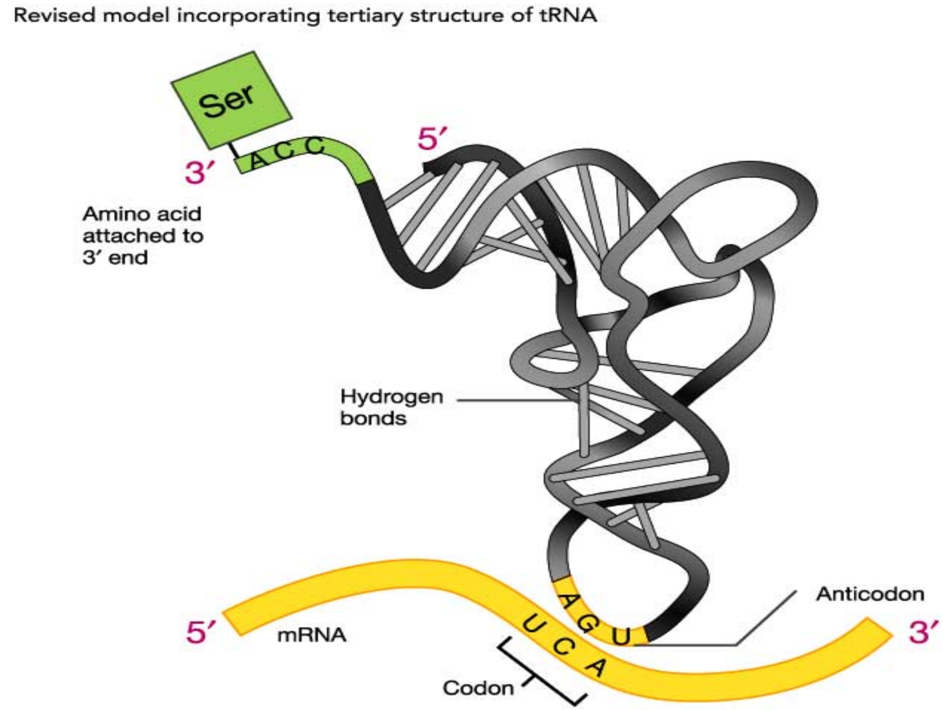
- If the string is “UCAUGAUCUCGUAAGA..”;



Anticodons & tRNA

Anticodon: is a unit made up of three nucleotides in t-RNA that correspond to the three bases of the codon on the mRNA and involve complementary binding with codon.



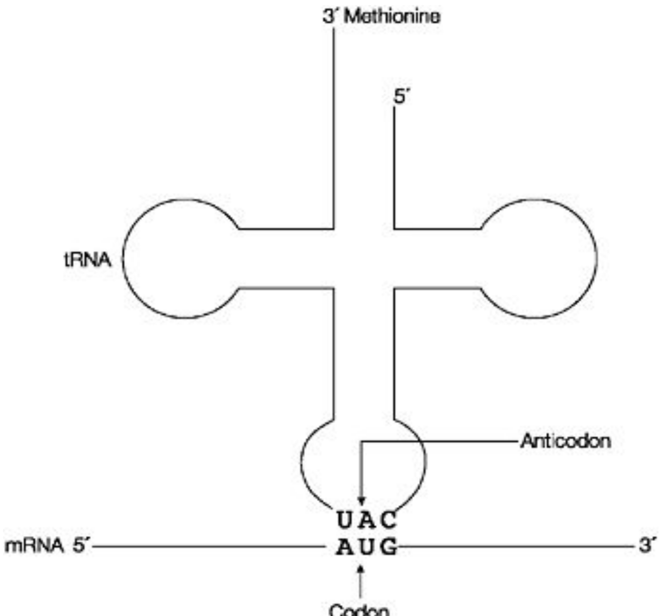


Conventional complementary pairing (A=U, C≡G) occur b/w first two bases of codon and the last two bases of anticodon. The third base of the codon is rather lenient or flexible with regard to the complementary base.

Wobble hypothesis:

The third base in the codon often fails to recognize the specific complementary base in the anticodon, b/c of this flexibility of the third base in the codon, single tRNA can recognize more than one codon.

Anticodon		Codon	
C	-	G	
A	-	U	Conventional base pairing
U	-	G or A	
G	-	U or C	Non-conventional base (Coloured) pairing



Wobble hypothesis:

How the Wobble Base of the Anticodon Determines the Number of Codons a tRNA Can Recognize

1. One codon recognized:

Anticodon	(3′) X–Y–**C** (5′)	(3′) X–Y–**A** (5′)
Codon	(5′) Y–X–**G** (3′)	(5′) Y–X–**U** (3′)

2. Two codons recognized:

Anticodon	(3′) X–Y–**U** (5′)	(3′) X–Y–**G** (5′)
Codon	(5′) Y–X–**A**/**G** (3′)	(5′) Y–X–**C**/**U** (3′)

3. Three codons recognized:

Anticodon	(3′) X–Y–**I** (5′)
Codon	(5′) Y–X–**A**/**U**/**C** (3′)

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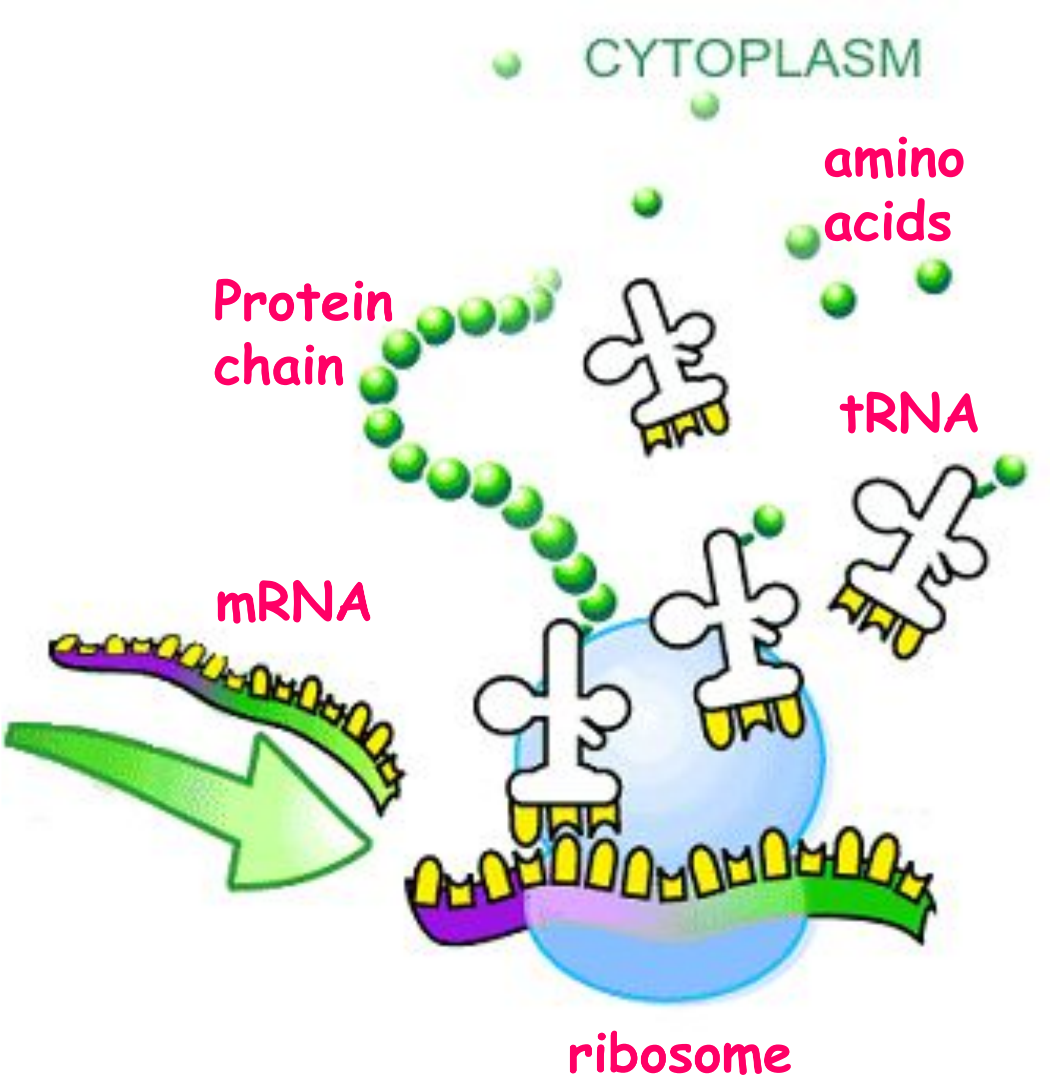
Translation – Overview

1. mRNA attaches to ribosome
2. tRNA brings in amino acids
3. Codon-anticodon match
4. Dehydration synthesis – protein chain elongates
5. Message done
    - Protein released
    - Ribosome falls apart
    - mRNA breaks up



Translation

- Translation - a multi-step process includes;
  1. Activation of amino acids
  2. Initiation
  3. Elongation
  4. Termination and release
  5. Folding and post-translational processing

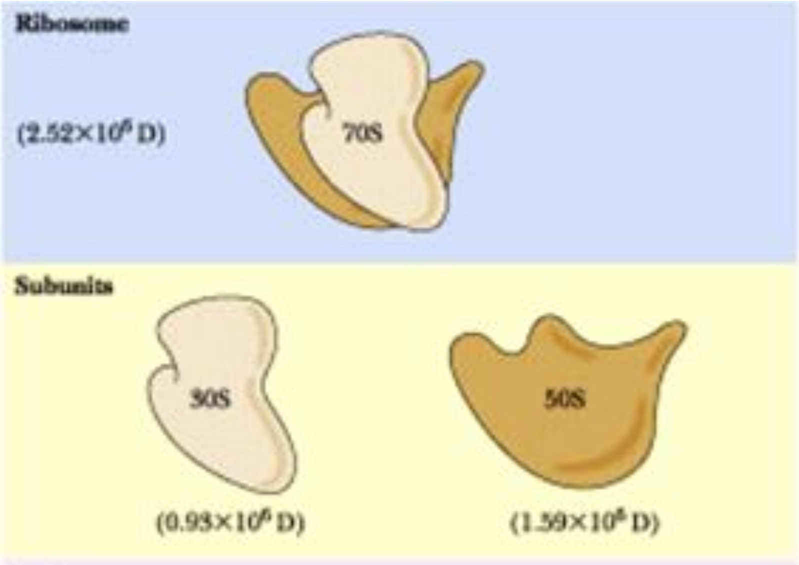
- Overall process - similar in both prokaryotes and eukaryotes, although particular differences exist.

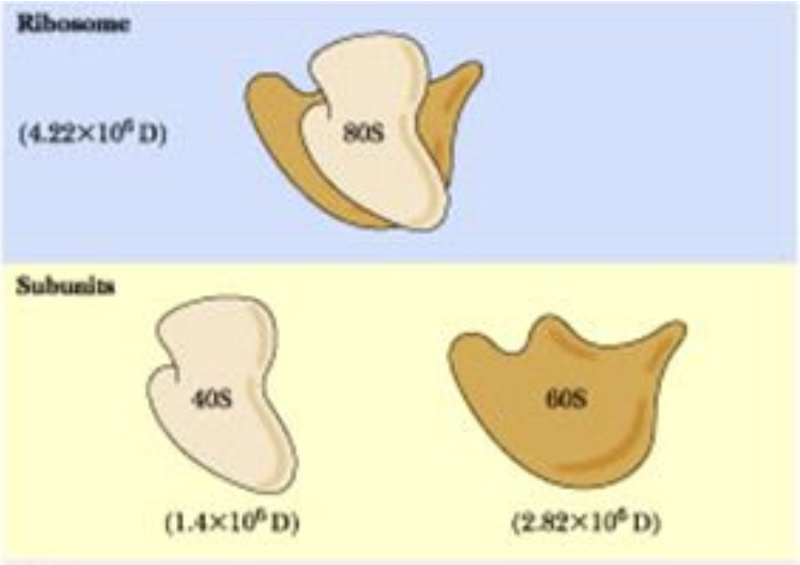
Essential components and Requirements in Bacteria

Stage	*Essential components*
1. Activation of amino acids	20 amino acids 20 aminoacyl-tRNA synthetases 32 or more tRNAs ATP Mg^{2+}
2. Initiation	mRNA *N*-Formylmethionyl-$tRNA^{fmet}$ Initiation codon in mRNA (AUG) 30S ribosomal subunit 50S ribosomal subunit Initiation factors (IF-1, IF-2, IF-3) GTP Mg^{2+}
3. Elongation	Functional 70S ribosome (initiation complex) Aminoacyl-tRNAs specified by codons Elongation factors (EF-Tu, EF-Ts, EF-G) GTP Mg^{2+}
4. Termination and release	Termination codon in mRNA Release factors (RF-1, RF-2, RF-3)
5. Folding and posttranslational processing	Specific enzymes, cofactors, and other components for removal of initiating residues and signal sequences, additional proteolytic processing, modification of terminal residues, and attachment of phosphate, methyl, carboxyl, carbohydrate, or prosthetic groups

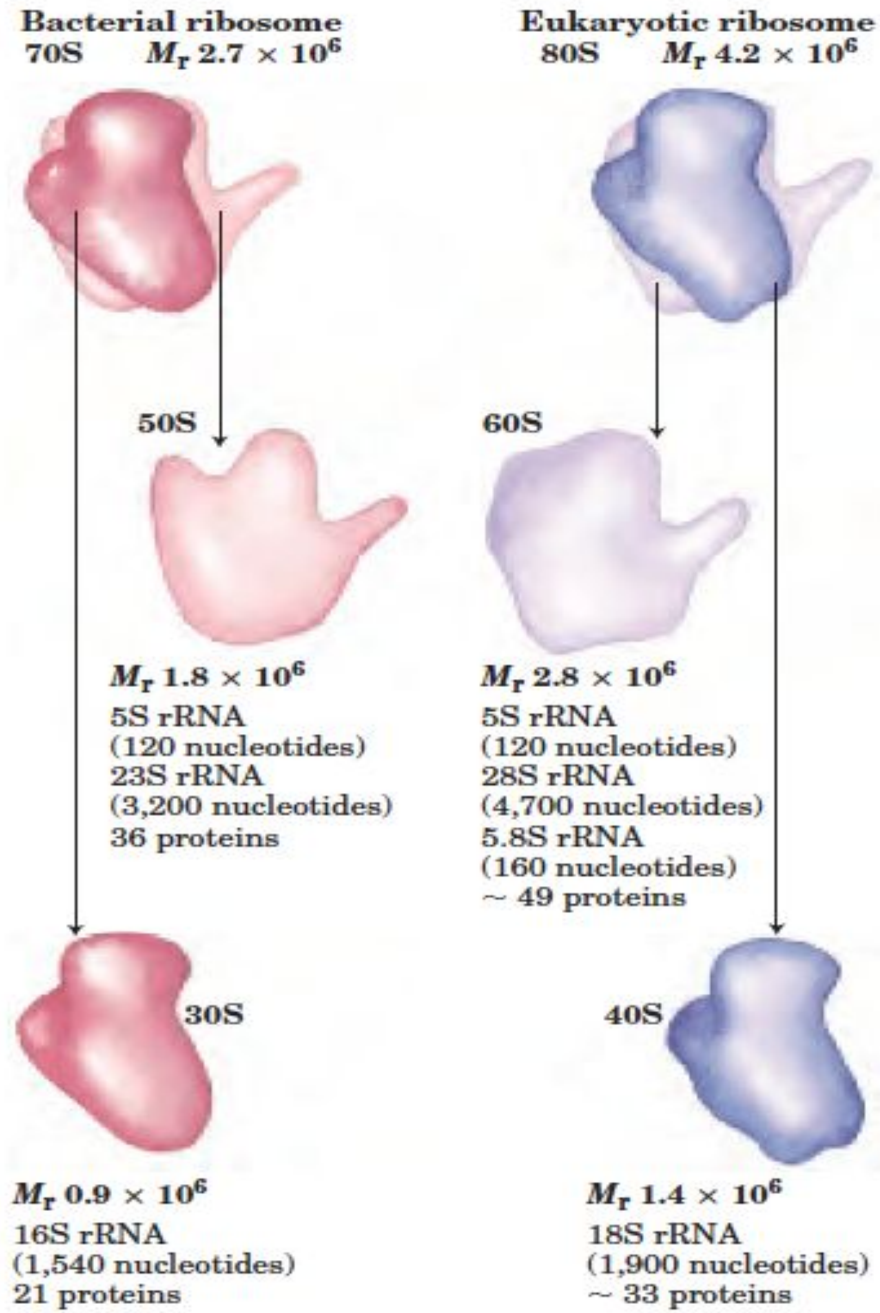
Prokaryotes vs. Eukaryotes

- Different sized ribosomal subunits





Prokaryotes vs. Eukaryotes



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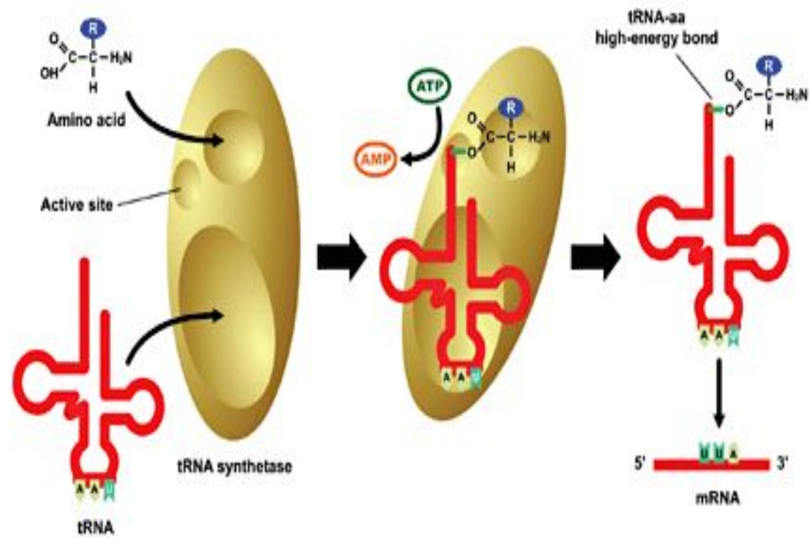
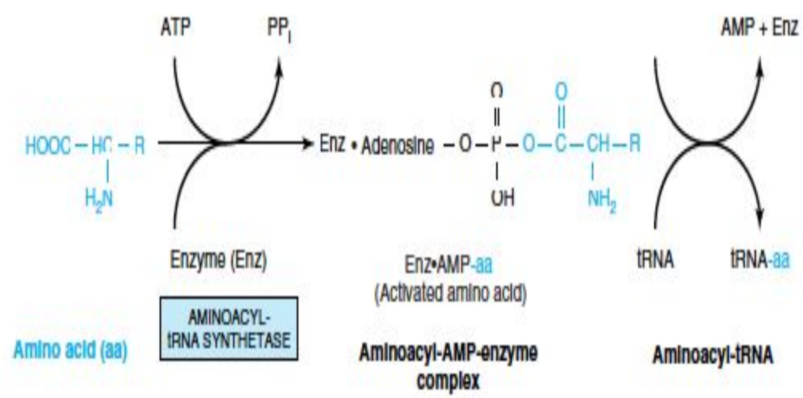
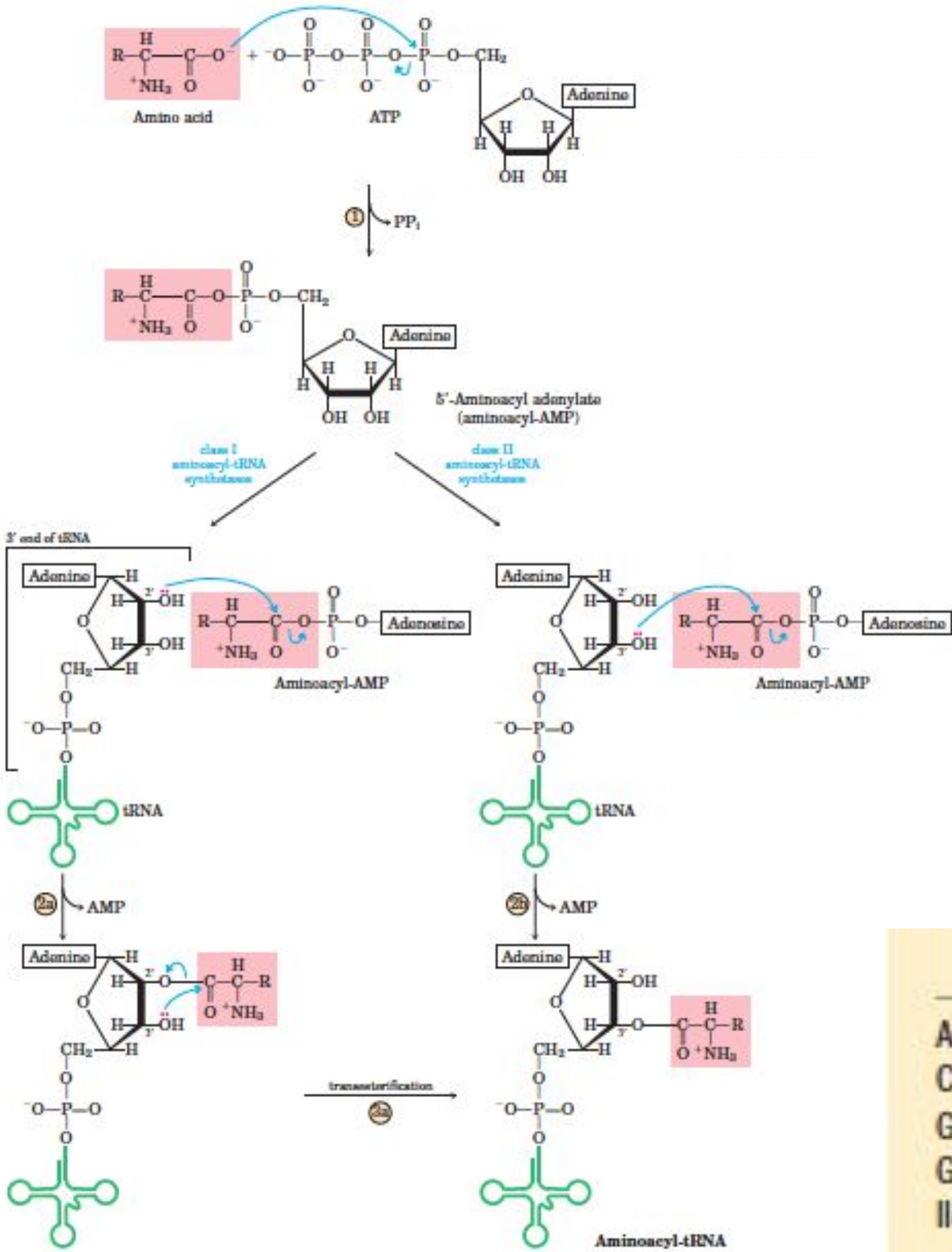
Stage 1. FORMATION OF AMINOACYL-tRNA

- A two-step reaction by aminoacyl-tRNA synthetase □ formation of aminoacyl-tRNA.
- The first reaction: Formation of an AMP-amino acid-enzyme complex.
- Next: The activated amino acid transferred to the corresponding tRNA molecule.
- The AMP and enzyme are released, and the latter can be reutilized.
- The charging reactions have an error rate of less than 10^{-4} and so are extremely accurate.





FORMATION OF AMINOACYL-tRNA



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The Two Classes of Aminoacyl-tRNA Synthetases

Class I		Class II	
Arg	Leu	Ala	Lys
Cys	Met	Asn	Phe
Gln	Trp	Asp	Pro
Glu	Tyr	Gly	Ser
Ile	Val	His	Thr

Stage 2. INITIATION

Initiation—the assembly of a ribosome on an mRNA

- The small subunit of the ribosome bound to initiator t-RNA scans the mRNA starting at the 5'end to identify and bind the initiation codon (AUG).
- Protein factors as well as sequences in mRNA are involved in the recognition of the initiation codon and formation of the initiation complex.
- The large subunit of the ribosome joins the small ribosomal subunit to generate the initiation complex at the initiation codon.

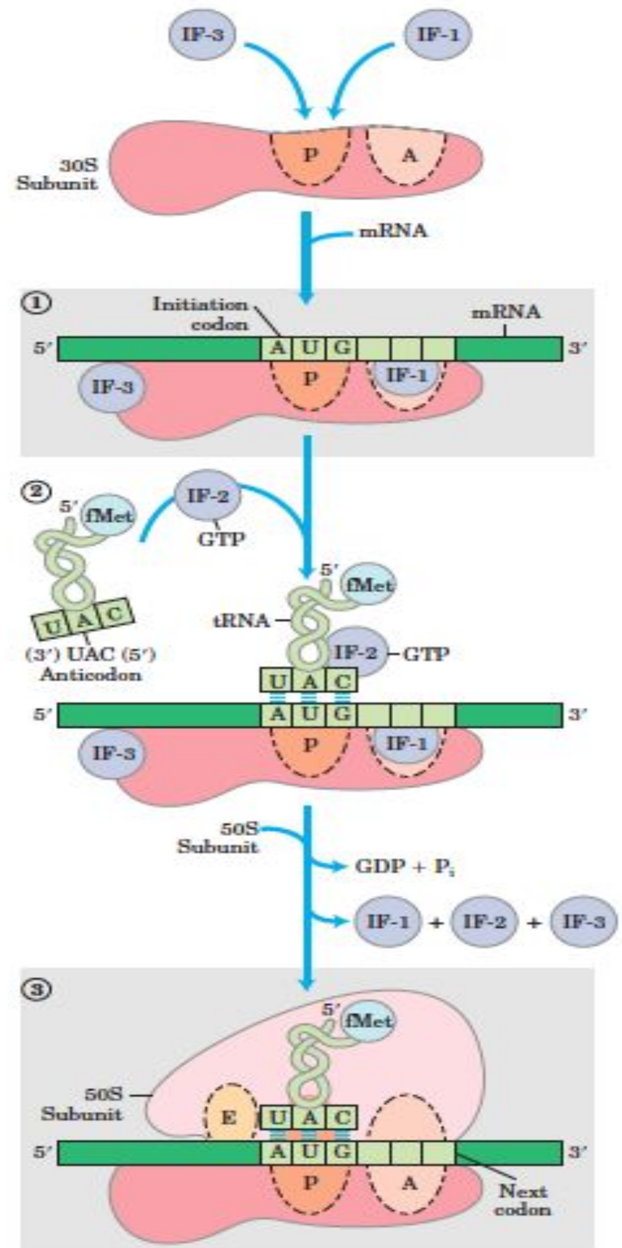
- Requirements: (In prokaryotes)
    - Large and small ribosome subunits
    - mRNA
    - Initiator tRNA
    - Three initiation factor(IFs) and GTP.

Protein Factors Required for Initiation of Translation in Bacterial and Eukaryotic Cells

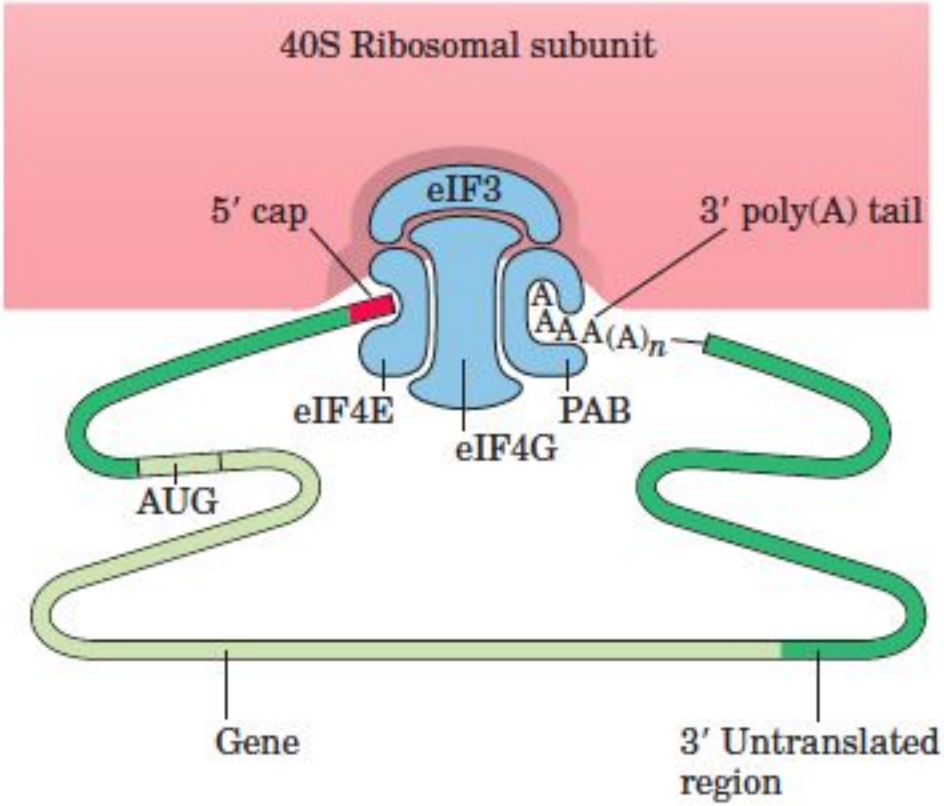
Factor	*Function*
Bacterial	
IF-1	Prevents premature binding of tRNAs to A site
IF-2	Facilitates binding of fMet-$tRNA^{fMet}$ to 30S ribosomal subunit
IF-3	Binds to 30S subunit; prevents premature association of 50S subunit; enhances specificity of P site for fMet-$tRNA^{fMet}$
Eukaryotic*	
eIF2	Facilitates binding of initiating Met-$tRNA^{Met}$ to 40S ribosomal subunit
eIF2B, eIF3	First factors to bind 40S subunit; facilitate subsequent steps
eIF4A	RNA helicase activity removes secondary structure in the mRNA to permit binding to 40S subunit; part of the eIF4F complex
eIF4B	Binds to mRNA; facilitates scanning of mRNA to locate the first AUG
eIF4E	Binds to the 5′ cap of mRNA; part of the eIF4F complex
eIF4G	Binds to eIF4E and to poly(A) binding protein (PAB); part of the eIF4F complex
eIF5	Promotes dissociation of several other initiation factors from 40S subunit as a prelude to association of 60S subunit to form 80S initiation complex
eIF6	Facilitates dissociation of inactive 80S ribosome into 40S and 60S subunits

Steps of Initiation:

1. IF1 and IF3 bind to the 30S subunit and prevent the large subunit binding.
2. IF2+GTP can then bind and will help the initiator tRNA to bind later.
3. This small subunit complex can now attach to an mRNA via its ribosome-binding site.
4. The initiator tRNA can then base-pair with the AUG initiation codon which releases IF3 thus creating the 30S initiation complex.
5. The large subunit then binds, displacing IF1 and IF2+GDP, giving the 70S initiation complex which is the fully assembled ribosome at the correct position on the mRNA.



Protein complexes in the formation of a eukaryotic initiation complex.



Stage 3. ELONGATION

Elongation—repeated cycles of amino acid delivery, peptide bond formation and movement along the mRNA.

- Charged tRNAs shuttled to the ribosome where they are polymerized to form a peptide.
- The sequence of amino acids added to the growing peptide is dependent on the mRNA sequence of the transcript.

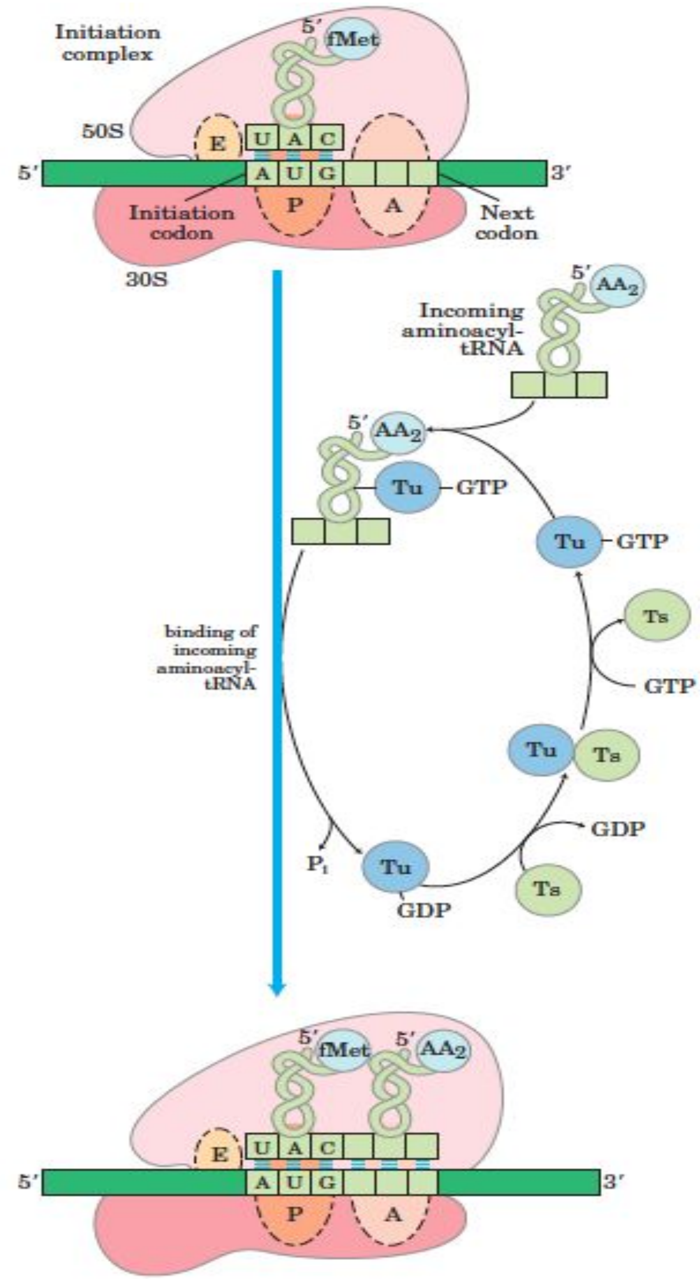
❖Requirements:

- Three factors (Efs), EF-Tu, EF-Ts and EF –G (in eukaryotes - eEF1α, eEF1βγ, and eEF2)
- GTP
- Charged tRNA and
- 70S initiation complex (or its equivalent).

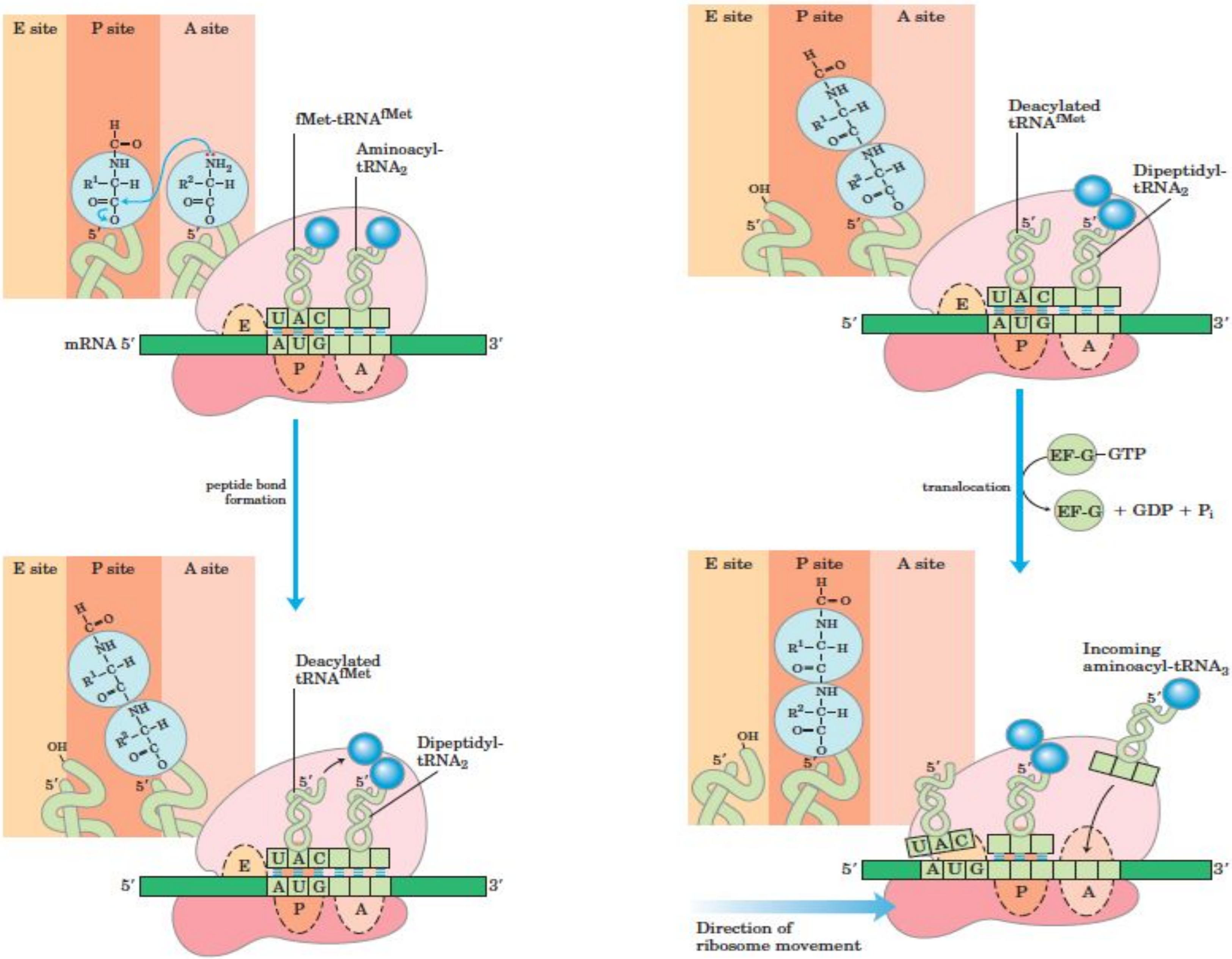
Steps of Elongation

- ❖ It takes place in three steps:

1. A charged tRNA is delivered as a complex with EF-Tu and GTP.
   - The GTP is hydrolyzed and EF-Tu.GDP is released which can be re-used with the help of EF-Ts and GTP (via the EF-Tu-EF-Ts exchange cycle).
2. Peptidyl transferase (23S rRNA) makes a peptide bond by joining the two adjacent amino acid without the input of more energy.
3. Translocase (EF-G), with energy from GTP, move the ribosome one codon along the mRNA, ejecting the uncharged tRNA and transferring the growing peptide chain to the P-site



Steps of Elongation



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Stage 4. TERMINATION

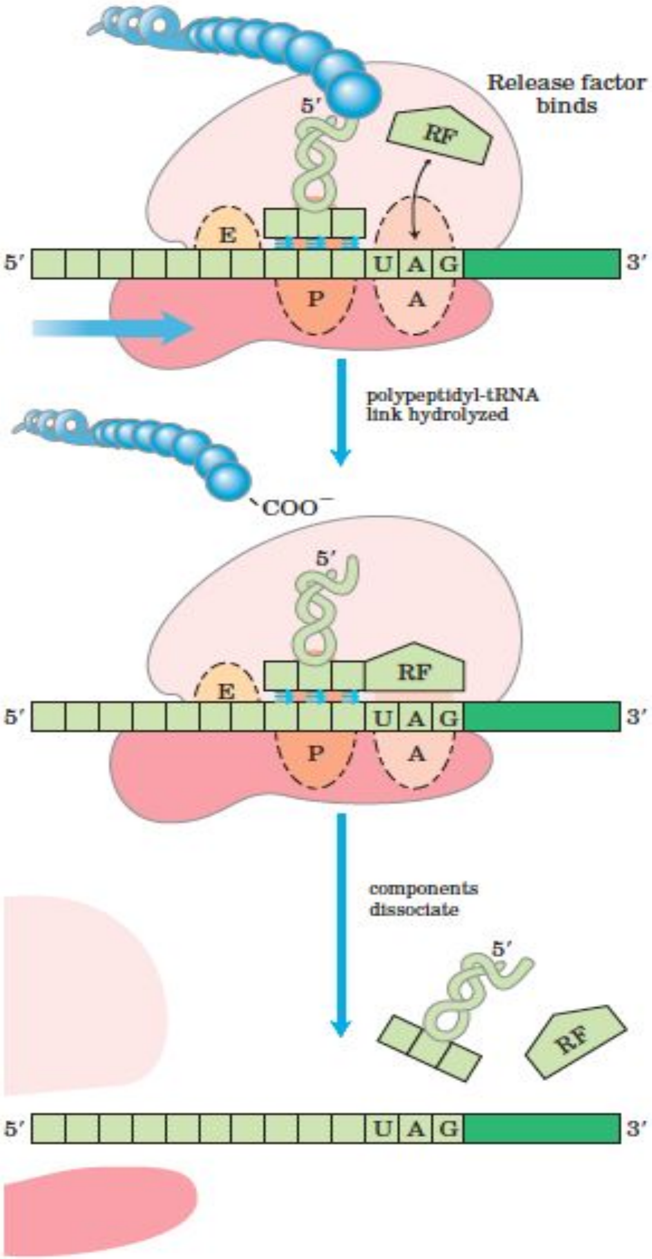
Termination—the release of the polypeptide chain.

- Finally, the nascent polypeptide is released when the ribosome reaches the termination codon (UAA, UAG & UGA).
- At this point, the ribosome is released from the mRNA and is ready to initiate another round of translation.
- Requirements:
  - Release factors: RF1, RF2 and RF3

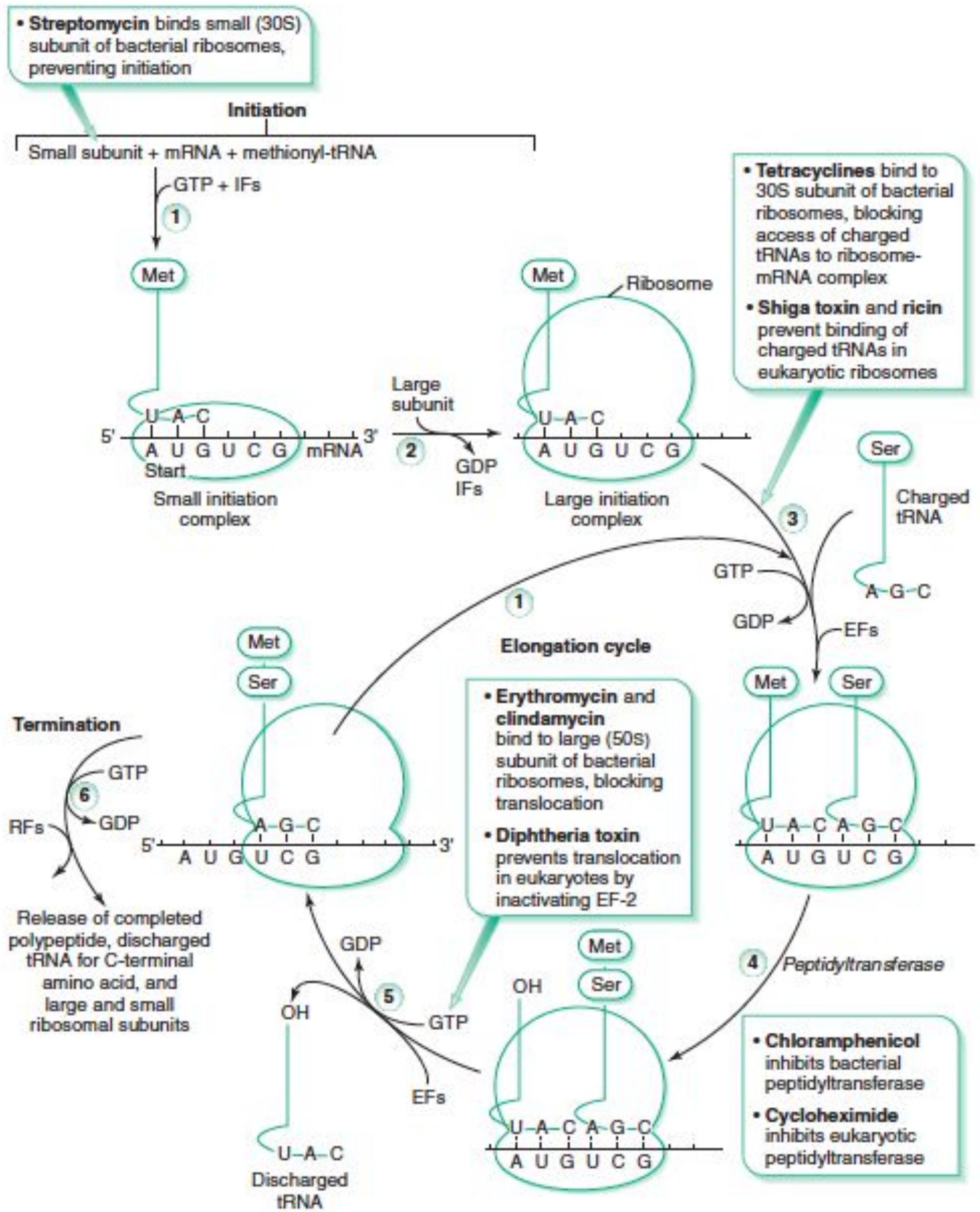
Steps of Termination:

- Release factors (RF1 or RF2) recognize the stop codon, helped by RF3, and make peptidyl transferase join the polypeptide chain to a water molecule, thus releasing it.

- Ribosome release factor helps to dissociate the ribosome subunit from the mRNA.



Summary of translation along with inhibitors

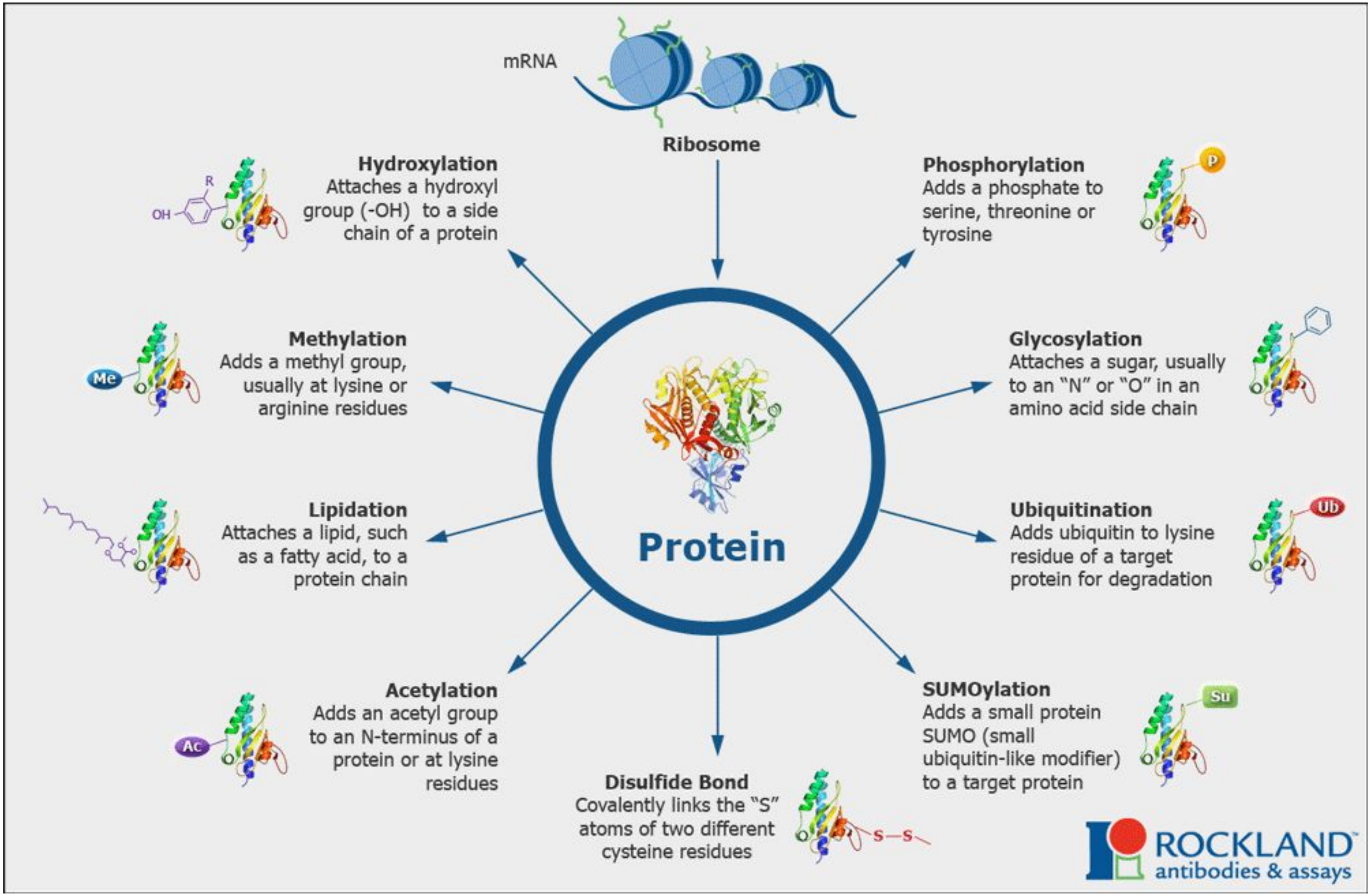


Stage 5: FOLDING AND POST-TRANSLATIONAL PROCESSING

- In order to achieve its biologically active form, the new polypeptide must fold into its proper three-dimensional conformation.

- Before or after folding, the new polypeptide may undergo enzymatic processing, including:
  - Removal of one or more amino acids (usually from the amino terminus);
  - Addition of acetyl, phosphoryl, methyl, carboxyl, or other groups to certain amino acid residues;
  - Proteolytic cleavage; and/or
  - Attachment of oligosaccharides or prosthetic groups.

Newly Synthesized Polypeptide Chains Undergo Folding and Processing



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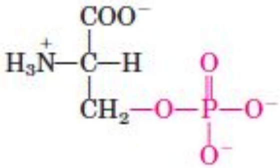
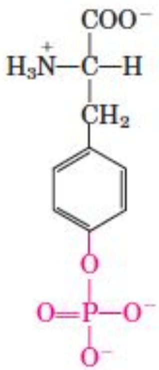
Posttranslational modifications

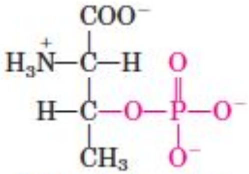
- Amino-Terminal and Carboxyl-Terminal Modifications
  - Removal of amino-terminal formyl/Met and additional residues (in some cases, carboxyl-terminal residues)
  - N-acetylation of amino-terminal residues (50% of eukaryotic proteins)

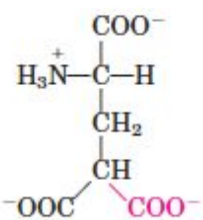
- Loss of Signal Sequences
  - 15 to 30 residues at the amino-terminal end (signal sequences) are ultimately removed by specific peptidases

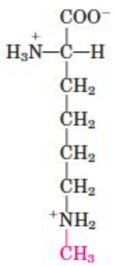
- Modification of Individual Amino Acids
  - Phosphorylation in casein (Ser, Thr, and Tyr)
  - Carboxylation in prothrombin (Glu)
  - Methylation in cytochrome c, muscle protein (Lys, glu)

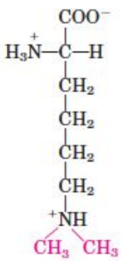

Phosphoserine

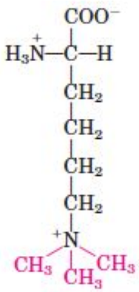

Phosphotyrosine

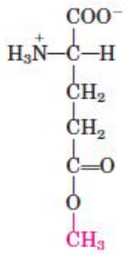

Phosphothreonine


γ-Carboxyglutamate


Methyllysine


Dimethyllysine


Trimethyllysine


Methylglutamate

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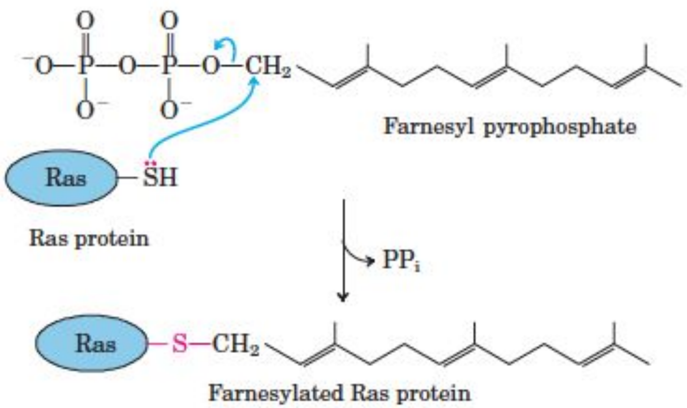
Posttranslational modifications

- Attachment of Carbohydrate Side Chains
  - Proteoglycans in membrane (Carbohydrate side chain attached covalently) (Asn,Ser, Thr)
- Addition of Isoprenyl Groups
  - Ras protein, G-protein, lamins
  - helps to anchor the protein in a membrane

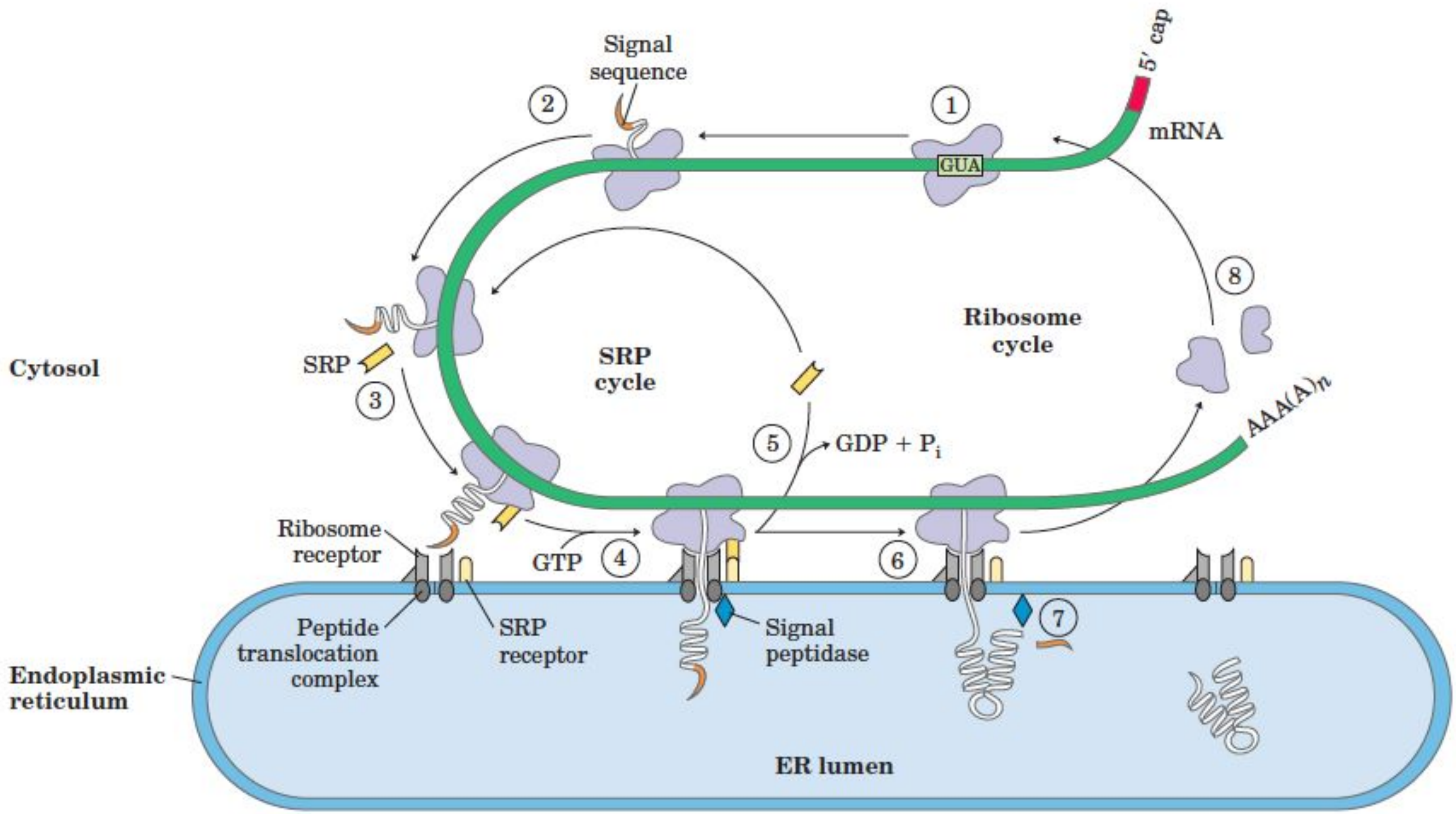


- Addition of Prosthetic Groups
  - Biotin of of acetyl-CoA carboxylase and the heme group of hemoglobin or cytochrome c
- Proteolytic Processing
  - Proteolytically trimmed to active forms (proinsulin, some viral proteins, and proteases such as chymotrypsinogen and trypsinogen)
- Formation of Disulfide Cross-Links
  - some proteins form intrachain or interchain disulfide bridges between Cys residues (protect the native conformation)

Protein Targeting and Degradation

- Posttranslational Modifications and addition of signals for targeting of Many Eukaryotic Proteins Begins in the Endoplasmic Reticulum

<u>Directing eukaryotic proteins with the appropriate signals to the endoplasmic reticulum</u>



Directing eukaryotic proteins with the appropriate signals to the endoplasmic reticulum

1. Targeting begins with initiation of protein synthesis on free ribosomes.

2. The signal sequence appears early in the synthetic process, because it is at the amino terminus, which as we have seen is synthesized first.

3. As it emerges from the ribosome, the signal sequence—and the ribosome itself—are bound by the large signal recognition particle (SRP);
   - SRP then binds GTP and halts elongation of the polypeptide when it is about 70 amino acids long and the signal sequence has completely emerged from the ribosome.

4. The GTP-bound SRP now directs the ribosome (still bound to the mRNA) to GTP-bound SRP receptors in the cytosolic face of the ER;
   - the nascent polypeptide is delivered to a peptide translocation complex in the ER, which may interact directly with the ribosome.

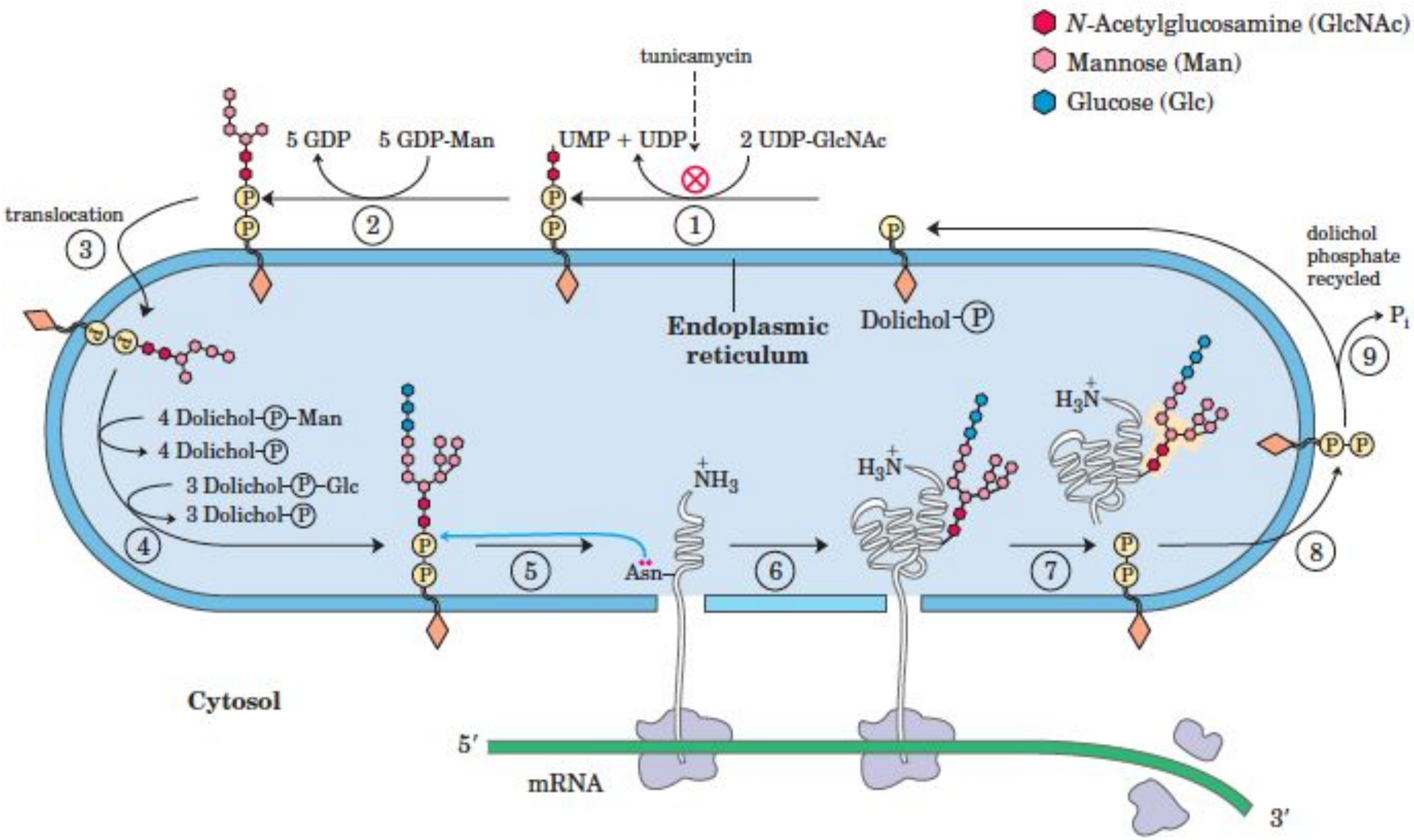
5. SRP dissociates from the ribosome, accompanied by hydrolysis of GTP in both SRP and the SRP receptor.

6. Elongation of the polypeptide now resumes, with the ATP-driven translocation complex feeding the growing polypeptide into the ER lumen until the complete protein has been synthesized.

7. The signal sequence is removed by a signal peptidase within the ER lumen;

8. the ribosome dissociates and is recycled.

Synthesis of the core oligosaccharide of glycoproteins.



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Synthesis of the core oligosaccharide of glycoproteins

1 , 2 steps occur on the cytosolic face of the ER.

3 Translocation moves the incomplete oligosaccharide across the membrane (mechanism not shown), and

4 completion of the core oligosaccharide occurs within the lumen of the ER.

- The precursors that contribute additional mannose and glucose residues to the growing oligosaccharide in the lumen are dolicho phosphate derivatives.

5 , 6 the core oligosaccharide is transferred from dolichol phosphate to an Asn residue of the protein within the ER lumen.

- The core oligosaccharide is then further modified in the ER and the Golgi complex in pathways that differ for different proteins.

- The five sugar residues shown surrounded by a beige screen (after step 7) are retained in the final structure of all N-linked oligosaccharides.

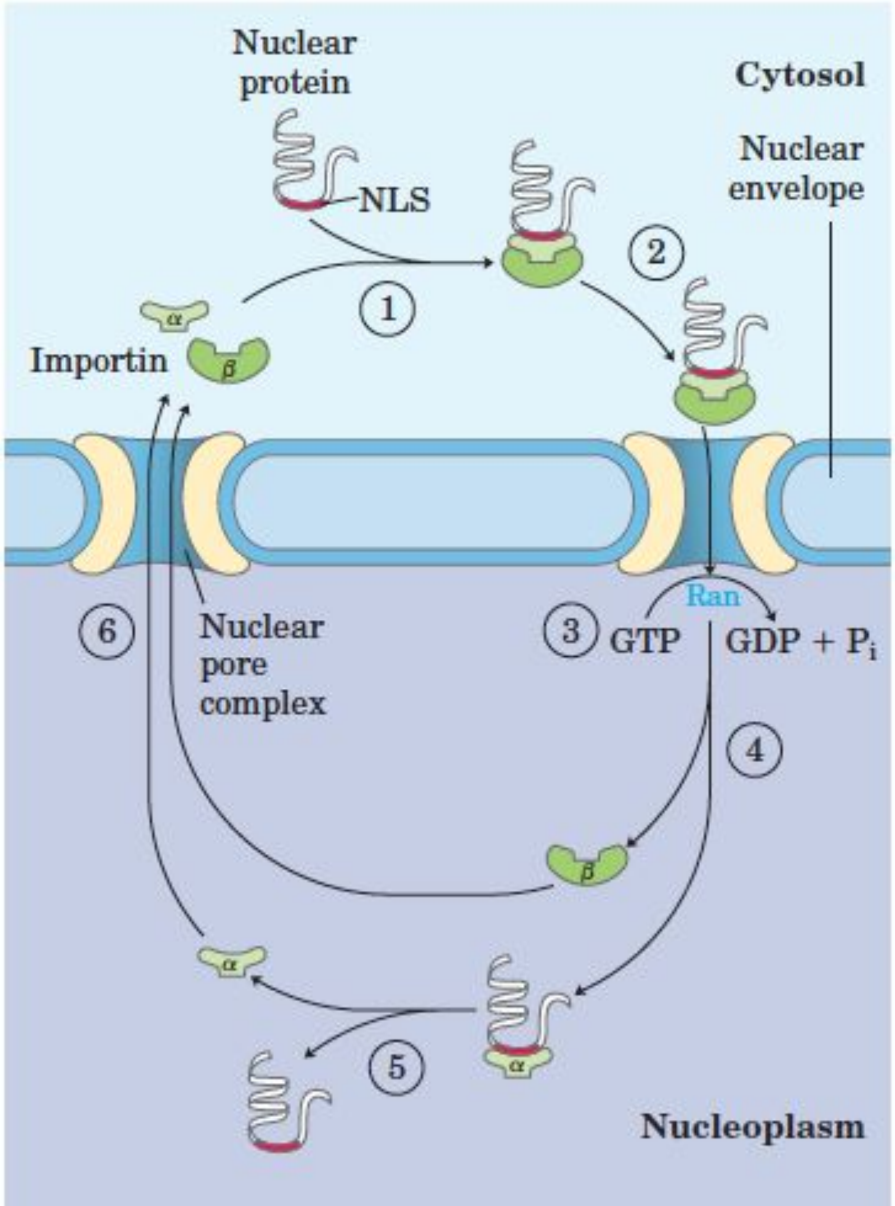
8 The released dolichol pyrophosphate is again translocated so that the pyrophosphate is on the cytosolic face of the ER, then

9 a phosphate is hydrolytically removed to regenerate dolichol phosphate.

Pathway taken by proteins destined for lysosomes, the plasma membrane, or secretion.

- Proteins are moved from the ER to the cis side of the Golgi complex in transport vesicles.
- Sorting occurs primarily in the trans side of the Golgi complex.

Targeting of nuclear proteins.



1. A protein with an appropriate nuclear localization signal (NLS) is bound by a complex of importin α and β.
2. The resulting complex binds to a nuclear pore, and
3. Translocation is mediated by the Ran GTPase.
4. Inside the nucleus, importin β dissociates from importin α, and
5. importin α then releases the nuclear protein.
6. Importin α and β are transported out of the nucleus and recycled.

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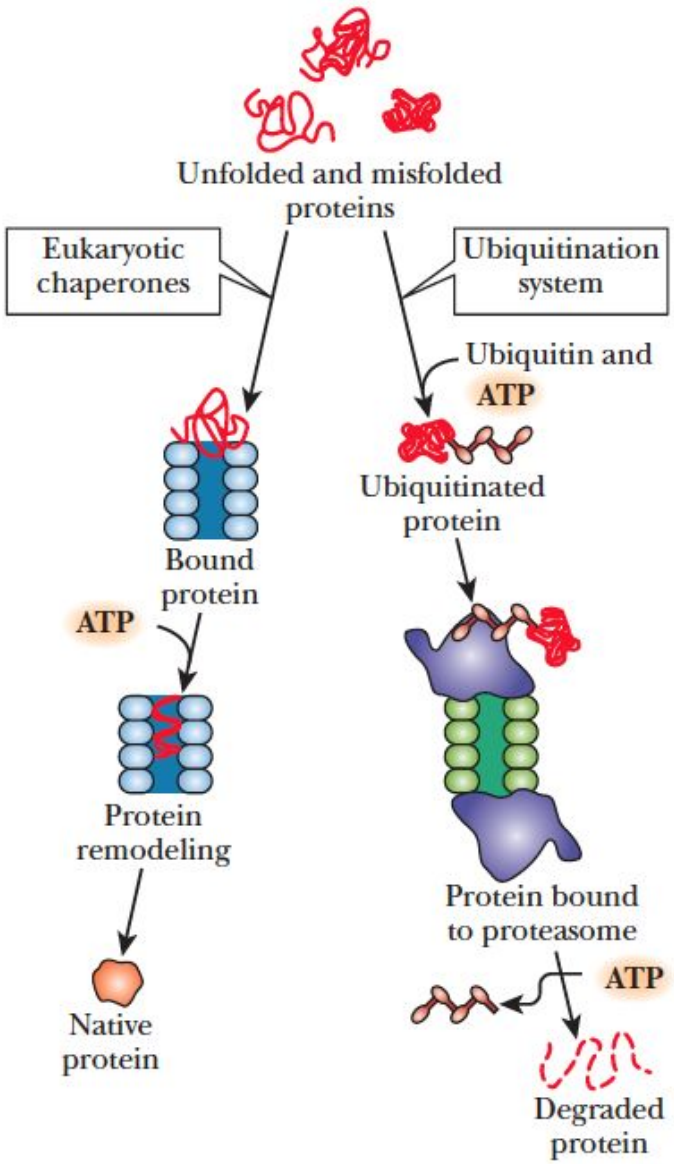
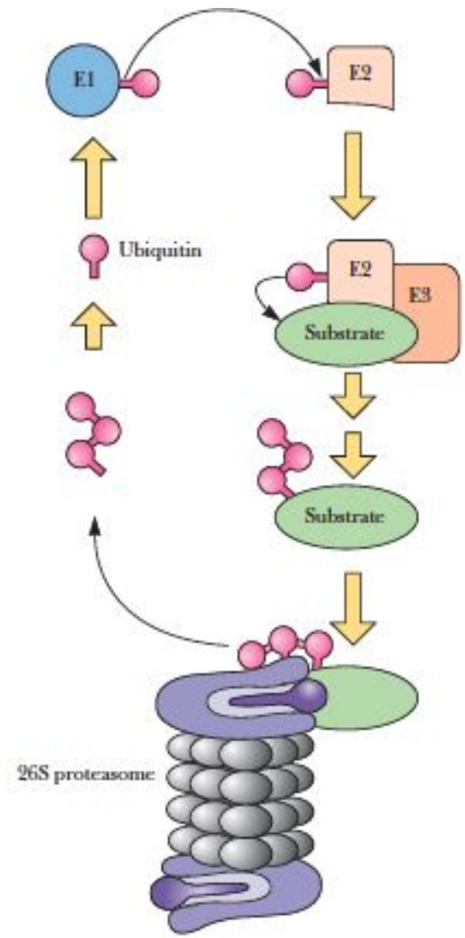
Folding and Degradation of proteins.





(Ubiquitin–proteasome degradation pathway).

(Misfolded proteins can be repaired and folded correctly by eukaryotic chaperones or they can be targeted for destruction by ubiquitination).

Similarities of prokaryotic and eukaryotic translation:

@. Both groups uses mRNA template

@. In both groups mRNA is synthesized from the genetic molecule, DNA

@. Ribosome is the protein synthesis machinery in both groups

@. All the 20 amino acids are same in both groups

@. All the 61 codons are similar in both groups

@. In both groups, protein synthesis occurs in the cytoplasm

@. In both groups, translation involves, (1). Initiation, (2). Elongation, (3). Termination

@. The process of peptide bond formation is similar in both groups

@. In both groups, tRNA pick up and bring the correct amino acid

@. In both groups the peptidyl transferase activity which catalyze the formation of peptide bond formation is done by rRNA molecules of larger sub-unit of ribosome (Ribozyme)

@. All the three stop codons are similar in both groups

Difference between prokaryotic and eukaryotic translation

Prokaryotic Translation

1. Process: Transcription and translation are continuous process and occurs simultaneously in the cytoplasm

2. Starting: 5’ end of mRNA is immediately available for translation

3. Ribosome: 70S type

4. Ribosome sub-units: 50S larger sub-unit and 30S smaller sub-unit

5. rRNA of larger ribosome sub-unit: Two rRNA molecules 5S and 23S rRNA

Eukaryotic Translation

1. Process: Transcription and translation are separate process, transcription occurs in the nucleus whereas translation occurs in the cytoplasm

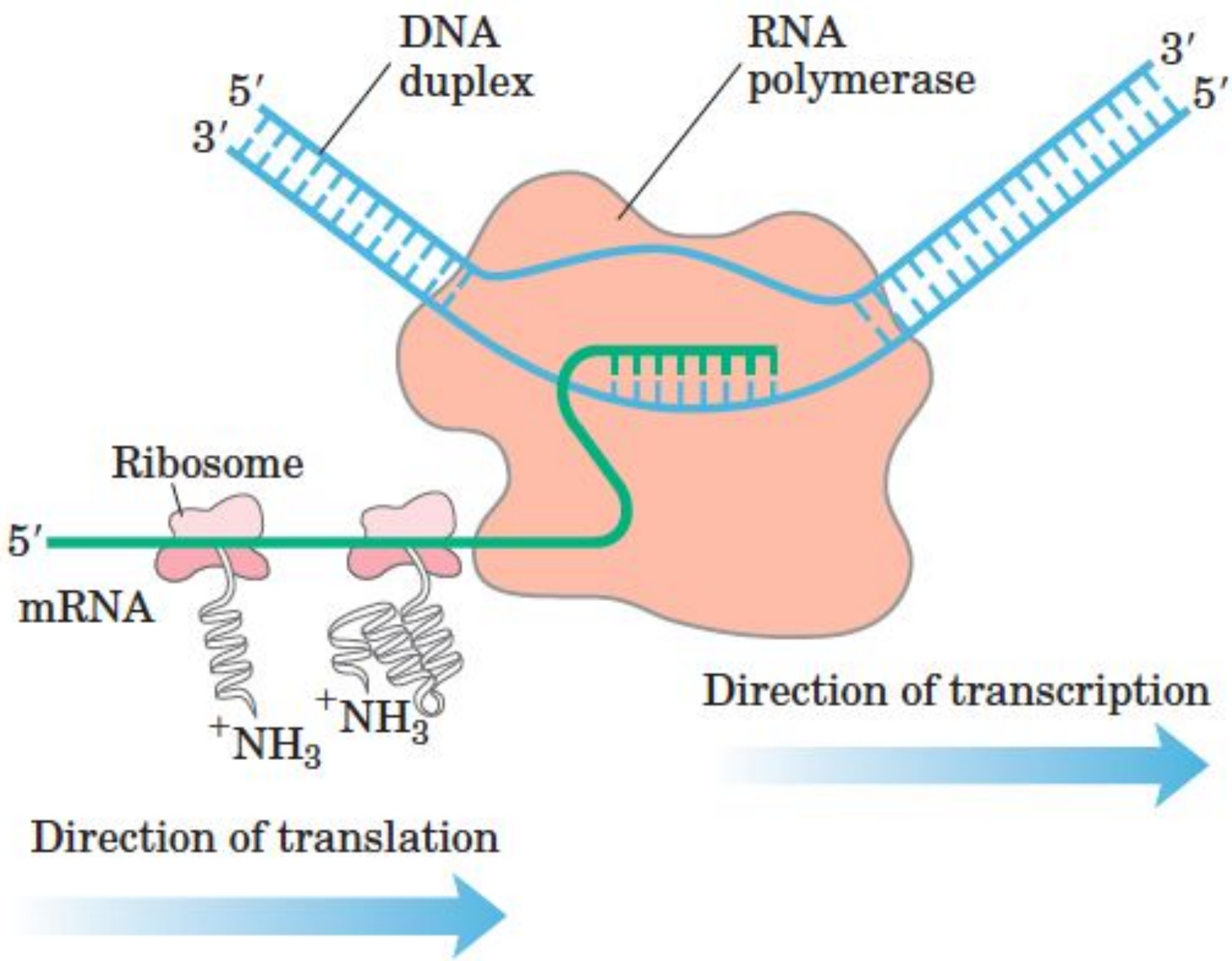
2. Starting: The primary transcript is processed after transcription and then it is transported to the cytoplasm, then only the cytoplasmic ribosomes can initiate translation

3. Ribosome: 80S type

4. Ribosome sub-units: composed of 60S larger subunit and 40S smaller subunit

5. rRNA of larger ribosome sub-unit: Three rRNA molecules, 5S, 5.8S and 28S rRNA

Coupling of transcription and translation in bacteria.



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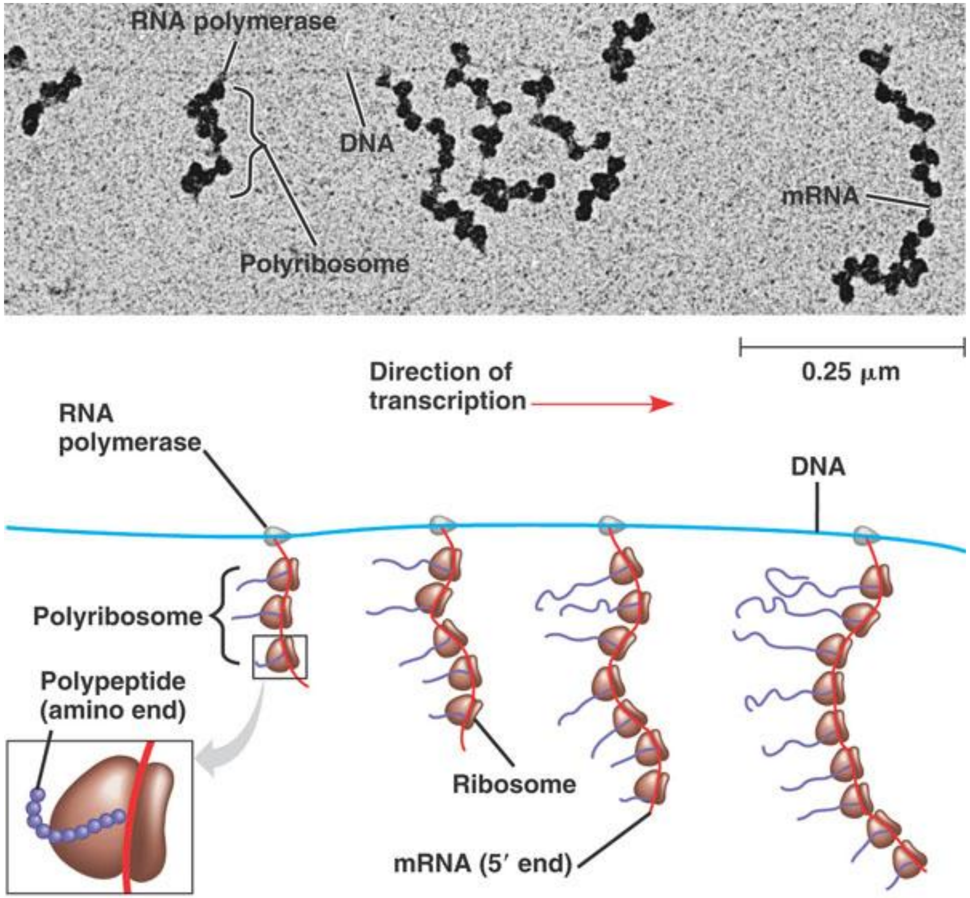
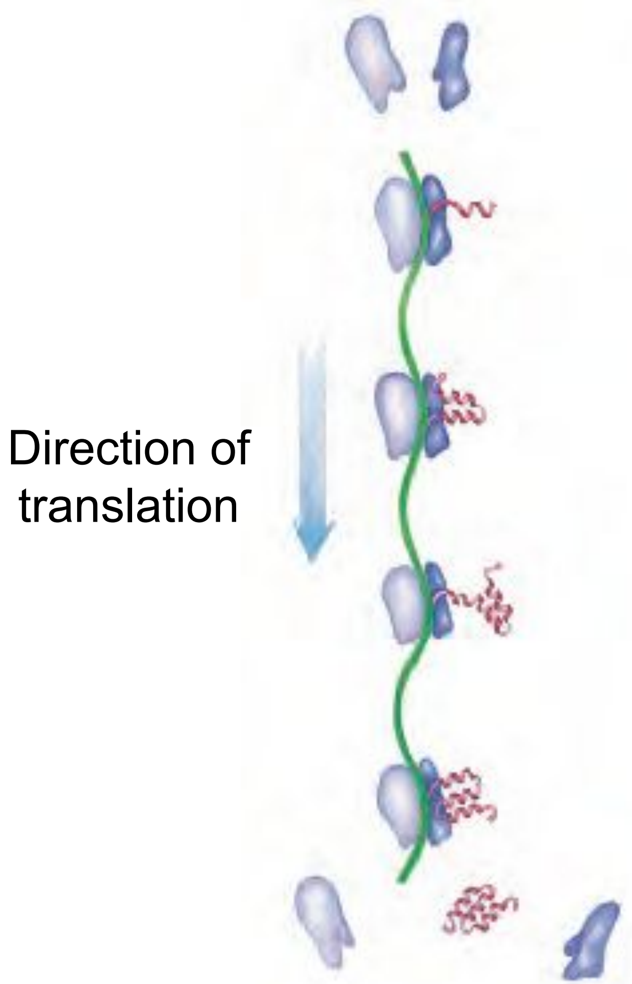
Prokaryotes vs. Eukaryotes

- Coupled T & T in prokaryotes but not in eukaryotes.



Rapid Translation of a Single Message by Polysomes



- Four ribosomes translating a eukaryotic mRNA molecule simultaneously, moving from the 5 end to the 3 end and synthesizing a polypeptide from the amino terminus to the carboxyl terminus.

Difference between prokaryotic and eukaryotic translation

Prokaryotic Translation	Eukaryotic Translation
6. rRNA of smaller ribosome sub-unit: One type, 16S rRNA	6. rRNA of smaller ribosome sub-units: 18S rRNA
7. Smaller ribosome sub-unit proteins: 21 proteins	7. Smaller ribosome sub-unit proteins: ~ 33 proteins
8. Larger ribosome sub-unit proteins: 36 proteins	8. Larger ribosome sub-unit proteins: ~49 proteins
9. Ribosome mass: 2700 kd	9. Ribosome mass: 4200 kd
10. Endoplasmic reticulum absent and hence protein synthesizing ribosome freely distributed in the cytoplasm	10. Endoplasmic reticulum present, protein synthesizing ribosome usually attached to the ER
11. mRNA is usually polycystronic	11. mRNA is always monocystronic
12. mRNA can acts as the template for the synthesis of many polypeptides	12. mRNA can act as the template for a single polypeptide

Difference between prokaryotic and eukaryotic translation

Prokaryotic Translation

13. Translation initiation: Single type which is cap independent initiation

14. Start site: May have many start sites and SD sequences (Shine-Dalgarno sequence) all along the mRNA

15. SD Sequence: SD sequence present 8 nucleotide upstream of start codon. SD sequence act as the ribosome binding site

16. Kozak sequence: Kozak sequence absent in mRMA

17. Initiation codon: Usually AUG, occasionally GUG or UUG

18. Smaller subunit of ribosome (30S) recognize the SD sequence in the mRNA during translation initiation

19. First tRNA: First tRNA is special type namely Met-tRNAf

Eukaryotic Translation

13. Translation initiation: Two types of translation initiation mechanisms. (1) Cap depended and (2) Cap independent

14. Start site: Always have only one start site which is located towards the 5’ region of mRNA

15. SD sequence: SD sequence is absent in mRNA

16. Kozak sequence present in the mRNA which is located few nucleotide upstream of start site. Kozak sequence assists initiation process of translation

17. Initiation codon: Initiation codon is AUG. occasionally GUG or CUG.

18. Smaller subunit of ribosome (40S) recognize the 5’ cap of mRNA during initiation

19. First tRNA: First tRNA is Met-tRNA

Difference between prokaryotic and eukaryotic translation

Prokaryotic Translation

20. First amino acid: First amino acid in the protein synthesis (methionine) will be formylated

21. Initiation factors: Only three initiation factors are required, they are IF1, IF2, IF3

22. Elongation factors: Two types of elongation factors, EF – Tu and EF – Ts

23. Speed of translation: ~ 20 amino acids/second

24. Termination: Facilitated by three release factors, RF1, RF2, RF3

25. Only the formyl group from the first amino acid (methionine) is removed from the polypeptide after protein synthesis

Eukaryotic Translation

20. First amino acid: No fomylation of methionine, the first amino acid, will occurs

21. Initiation factors: Seven types of initiation factors are required for translation, they are eIF1, eIF2, eIF3, eIF4, eIF5A, eIF5B, eIF6

22. Elongation factors: Elongation factors are eEF1 and eEF2

23. Speed of translation: ~1 amino acid/second

24. Termination: Termination is facilitated by only one release factor eRF1

25. Usually the un-formylated first methionine as such is removed from the polypeptide after protein synthesis

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Difference between prokaryotic and eukaryotic translation

Prokaryotic Translation

26. Life span of mRNA: Short, few seconds to few minutes

27. IF3 prevents the association of ribosomal subunits in the absence of initiation complex

28. Post translational modifications: Post translational modifications of proteins takes place in the cytoplasm

Eukaryotic Translation

26. Life span of mRNA: Life span of mRNA long, few hours to a day or sometimes more

27. eIF3 prevents the association of ribosomal subunits in the absence of initiation complex

28. Post translational modifications: Post translational modifications usually takes place in the endoplasmic reticulum or Golgi bodies or in the cytoplasm

Inhibitors of Protein & RNA Synthesis

INHIBITOR	SPECIFIC EFFECT
Acting only on bacteria	
Tetracycline	blocks binding of aminoacyl-tRNA to A-site of ribosome
Streptomycin	prevents the transition from initiation complex to chain-elongating ribosome and also causes miscoding
Chloramphenicol	blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6–65)
Erythromycin	blocks the translocation reaction on ribosomes (step 3 in Figure 6–65)
Rifamycin	blocks initiation of RNA chains by binding to RNA polymerase (prevents RNA synthesis)
Acting on bacteria and eucaryotes	
Puromycin	causes the premature release of nascent polypeptide chains by its addition to growing chain end
Actinomycin D	binds to DNA and blocks the movement of RNA polymerase (prevents RNA synthesis)
Acting on eucaryotes but not bacteria	
Cycloheximide	blocks the translocation reaction on ribosomes (step 3 in Figure 6–65)
Anisomycin	blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6–65)
α-Amanitin	blocks mRNA synthesis by binding preferentially to RNA polymerase II

The ribosomes of eucaryotic mitochondria (and chloroplasts) often resemble those of bacteria in their sensitivity to inhibitors. Therefore, some of these antibiotics can have a deleterious effect on human mitochondria.

Thank you………..

…………..Read More!

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